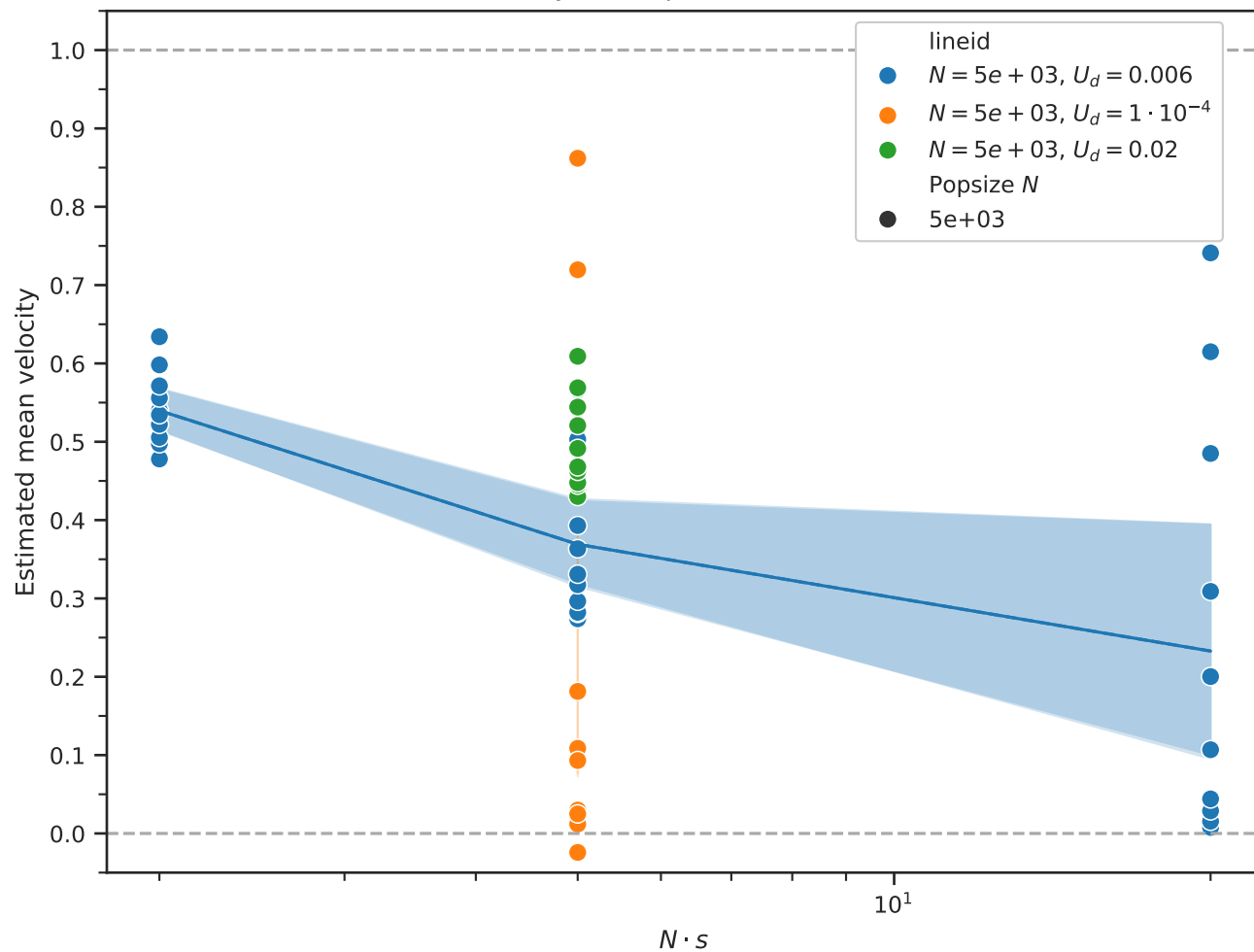
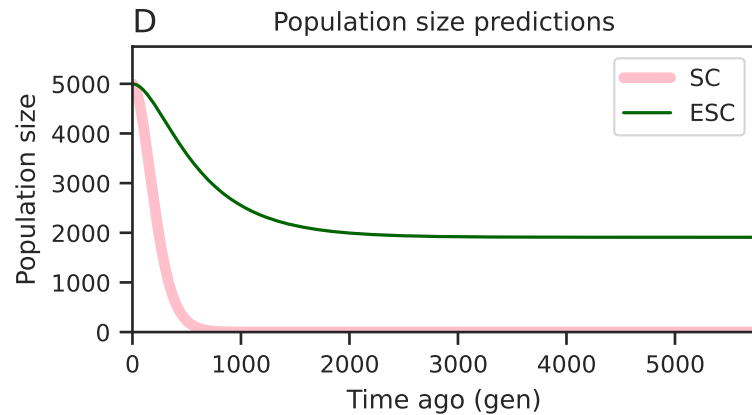
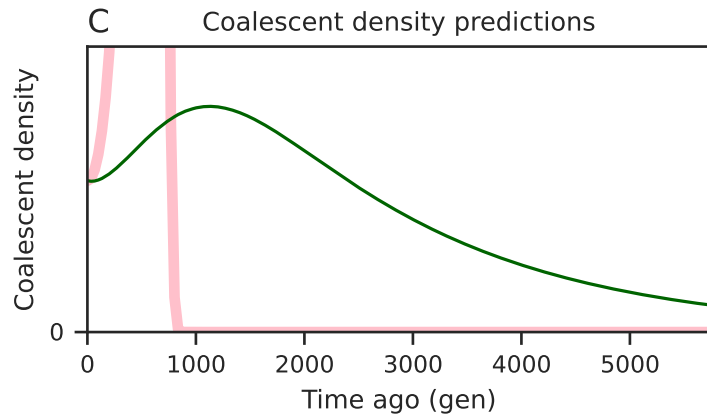
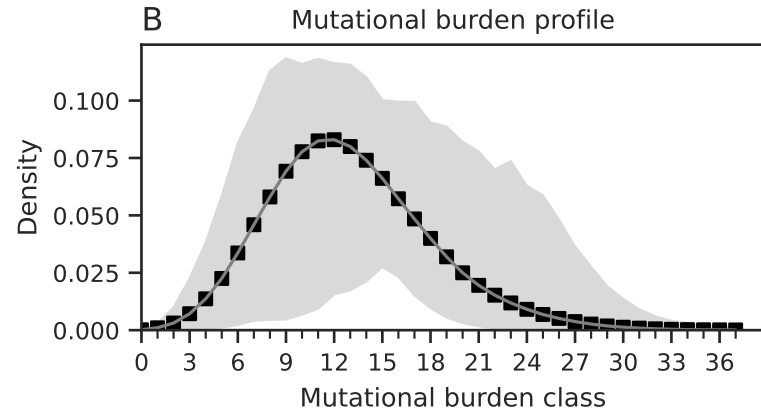
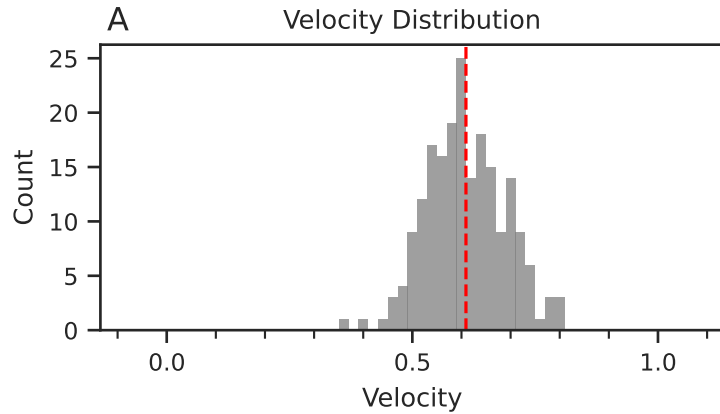


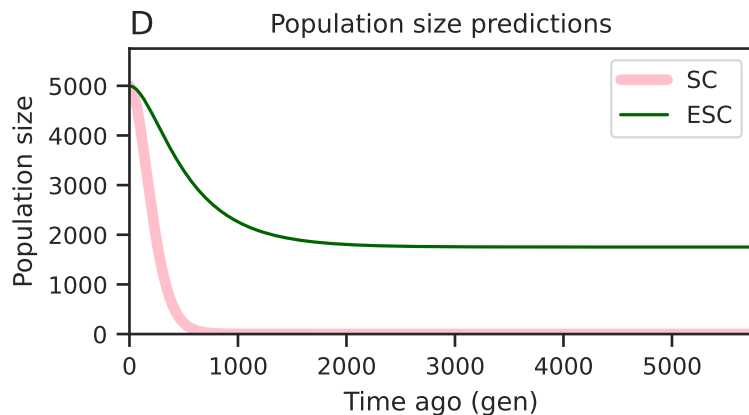
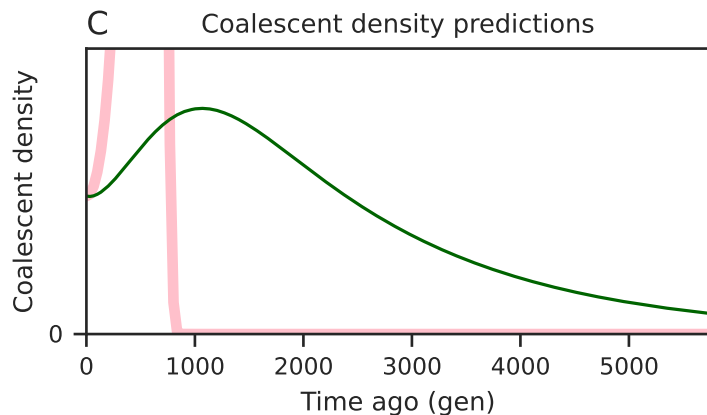
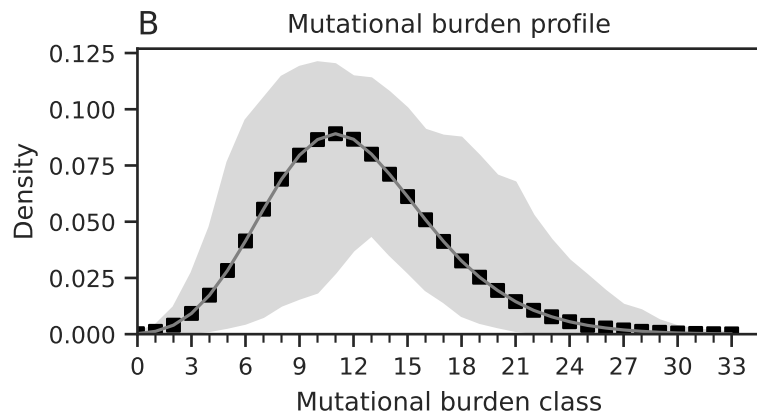
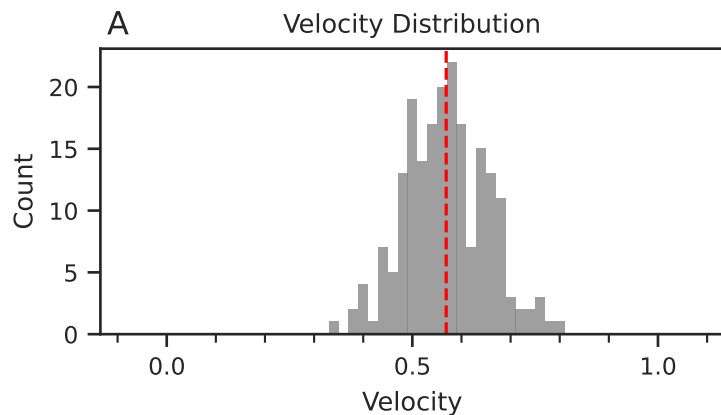
Mean velocity across parameter combinations



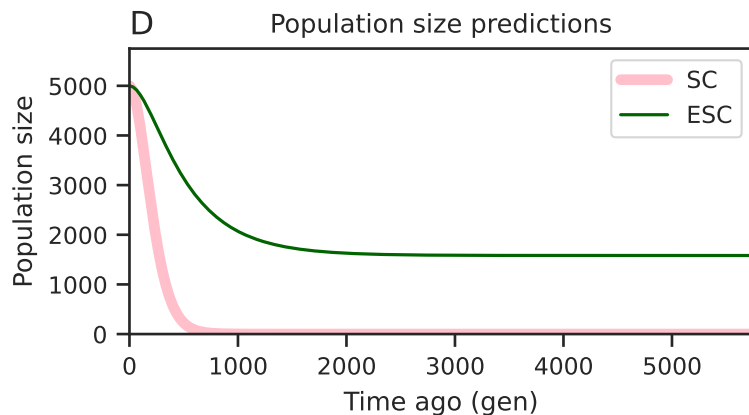
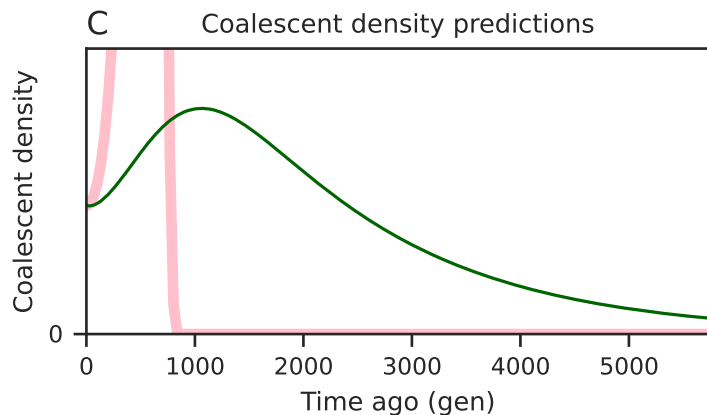
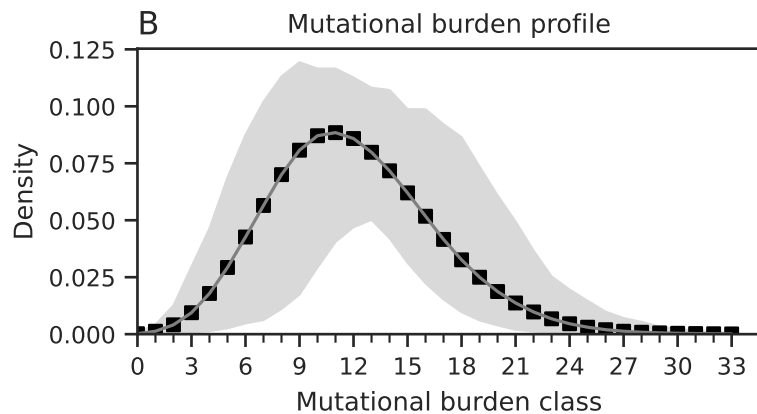
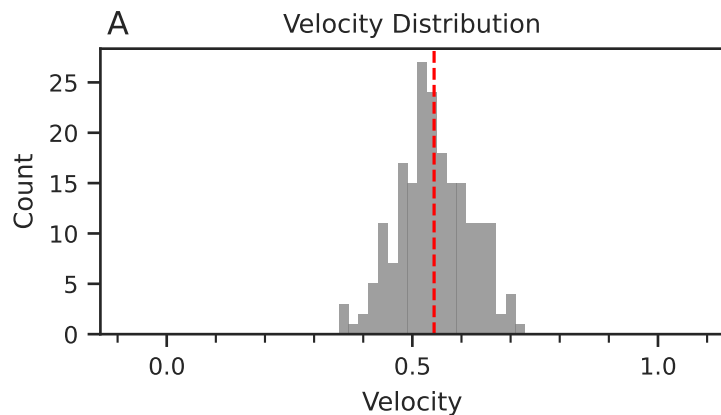
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 0.001$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.6091 ± 0.0793 , $n = 200$ simulations



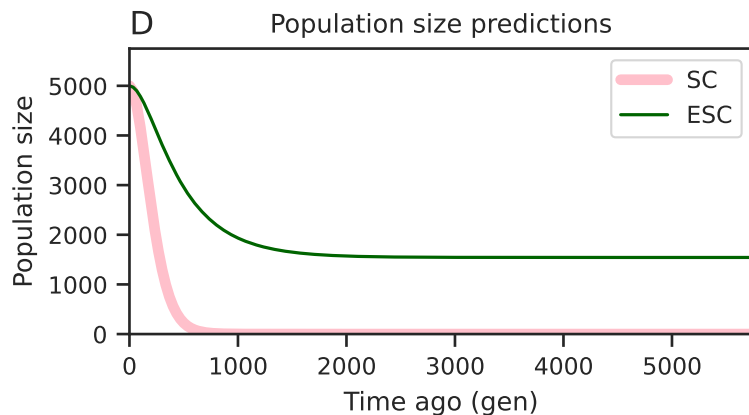
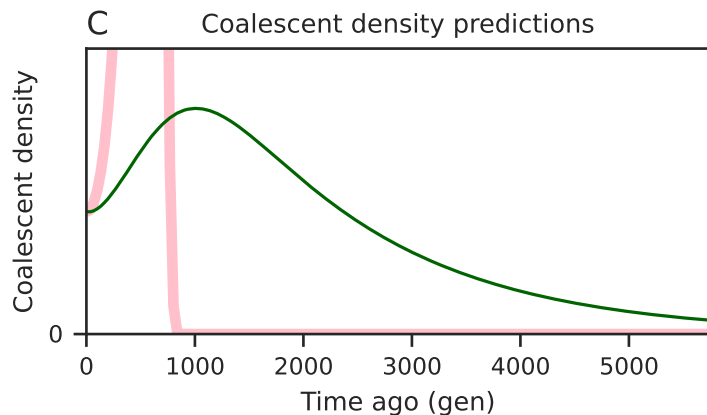
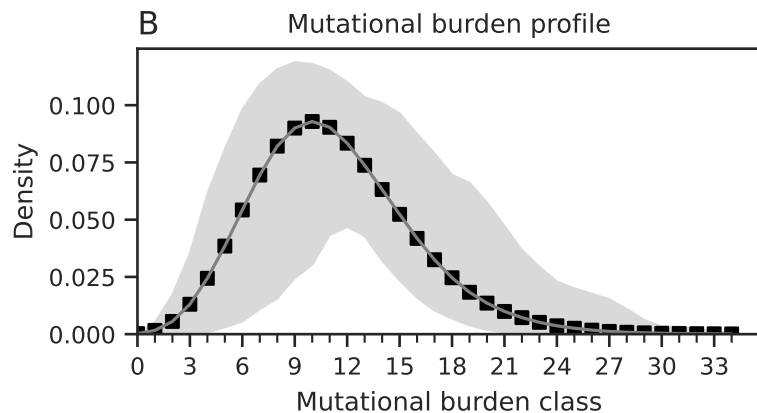
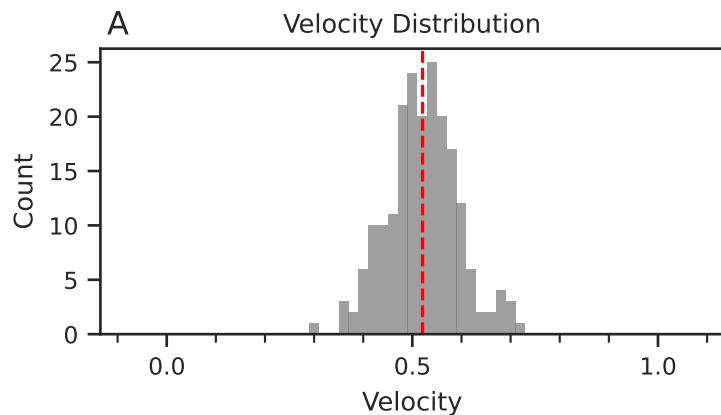
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 9 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.5689 ± 0.0828 , $n = 200$ simulations



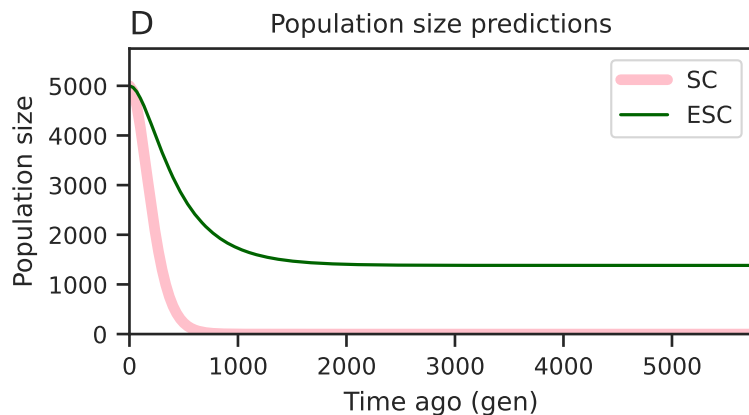
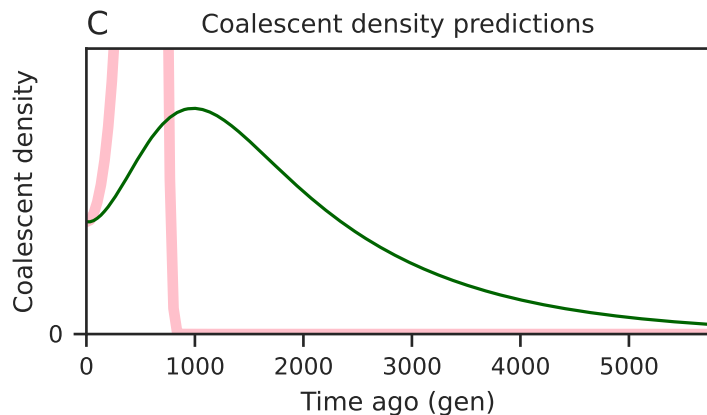
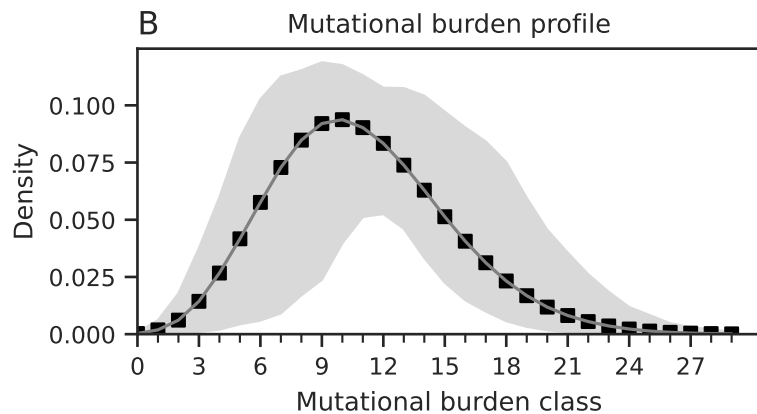
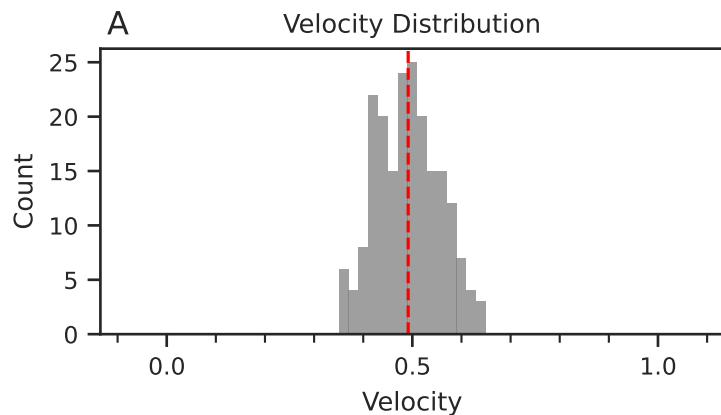
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 8 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.5443 ± 0.0730 , $n = 200$ simulations



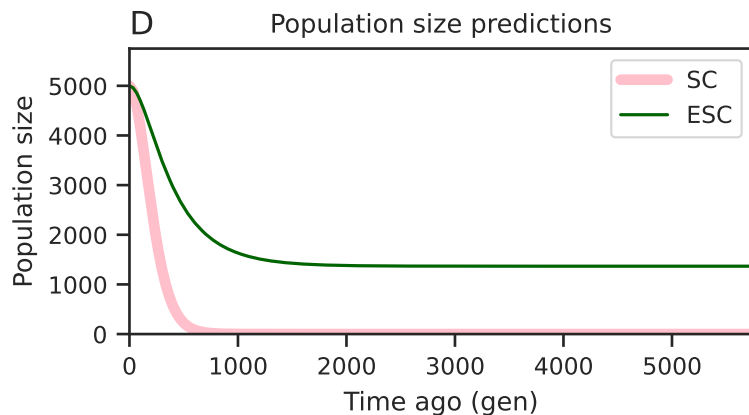
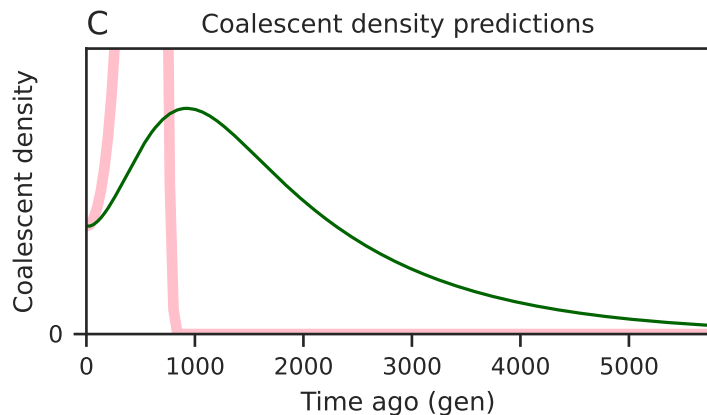
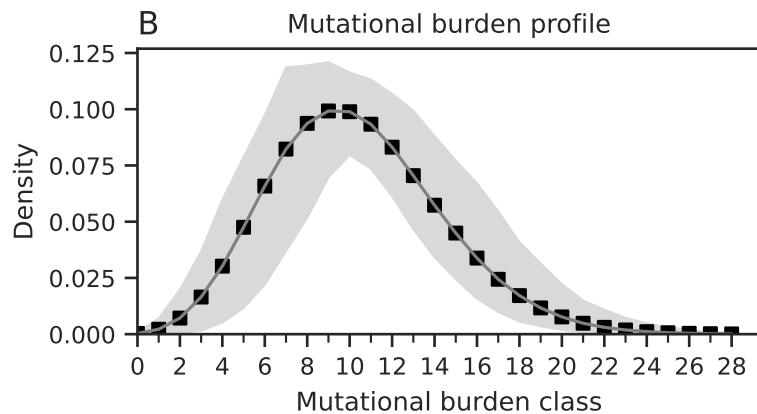
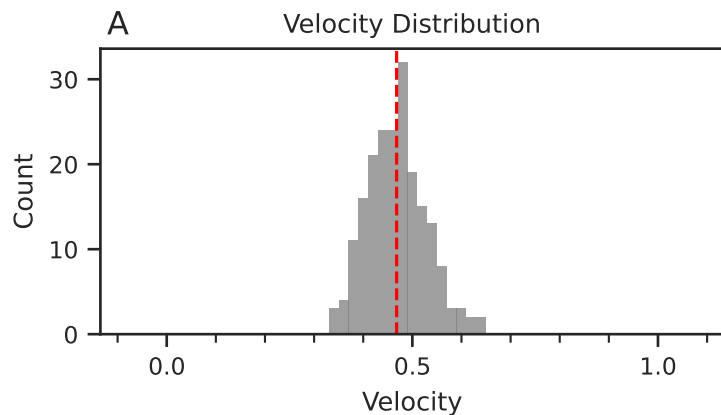
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 7 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.5208 ± 0.0716 , $n = 200$ simulations



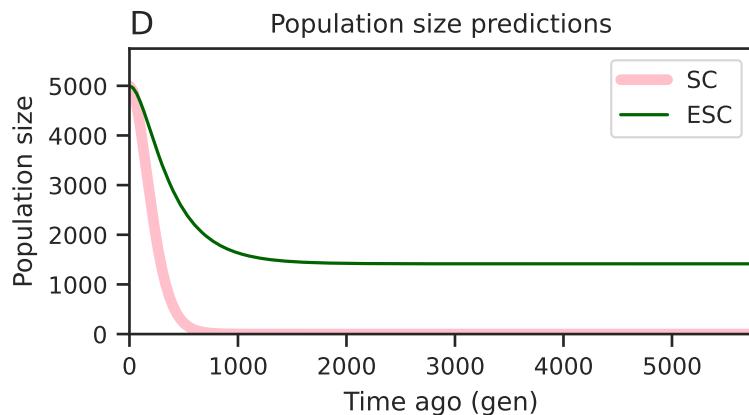
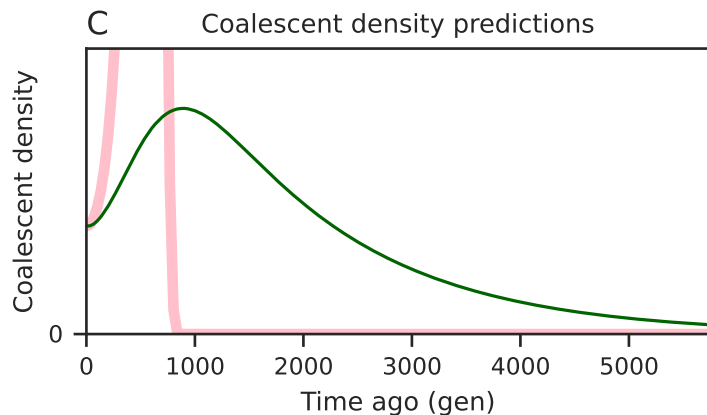
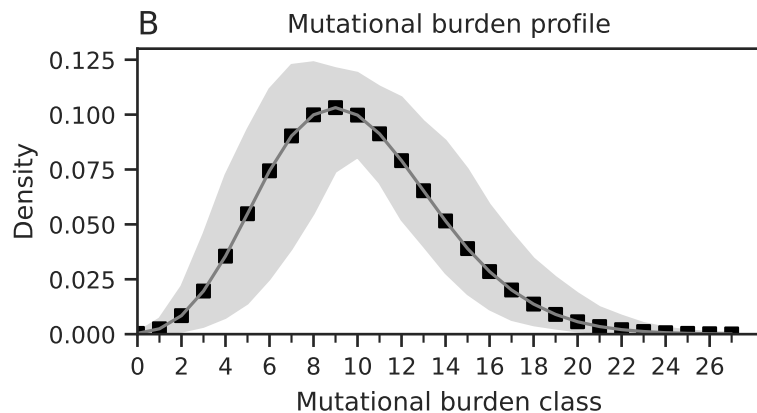
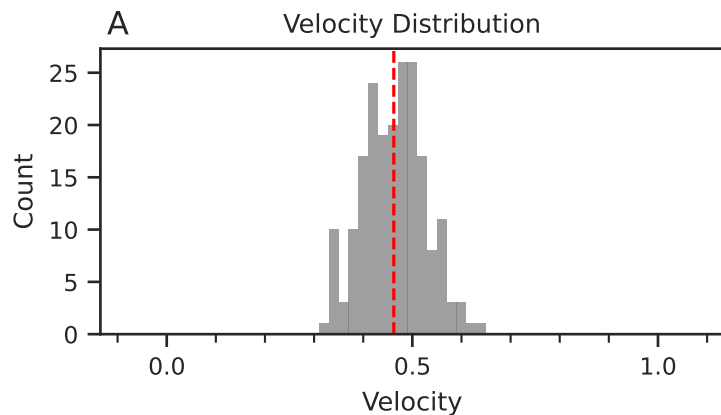
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 6 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4915 ± 0.0646 , $n = 200$ simulations



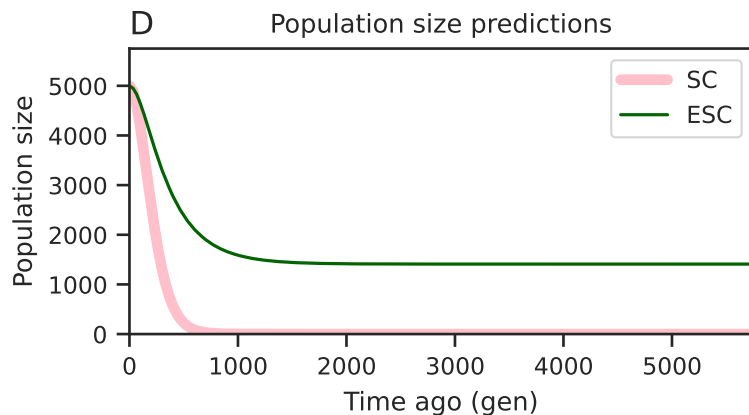
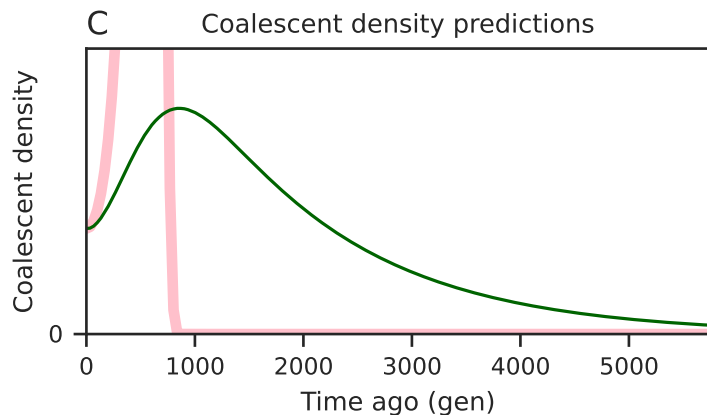
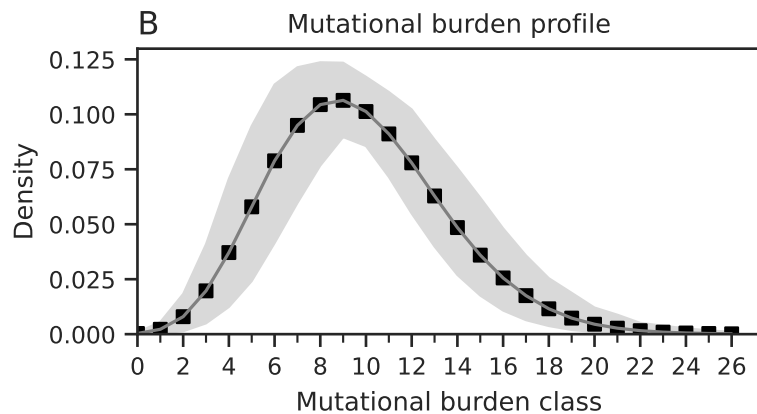
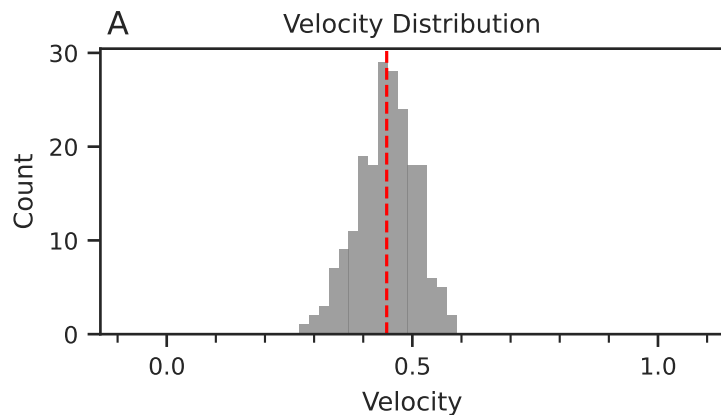
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4680 ± 0.0598 , $n = 200$ simulations



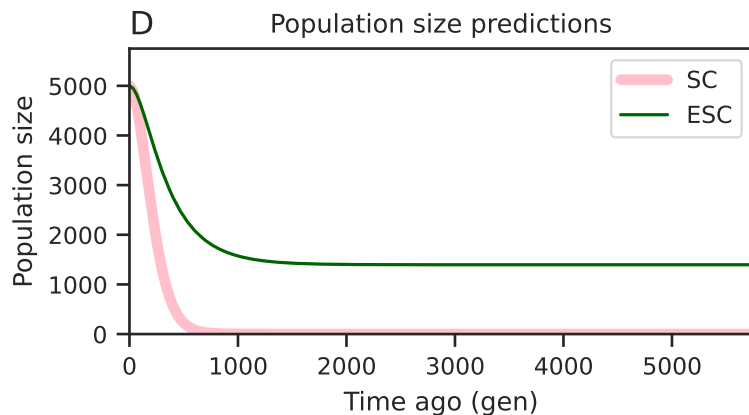
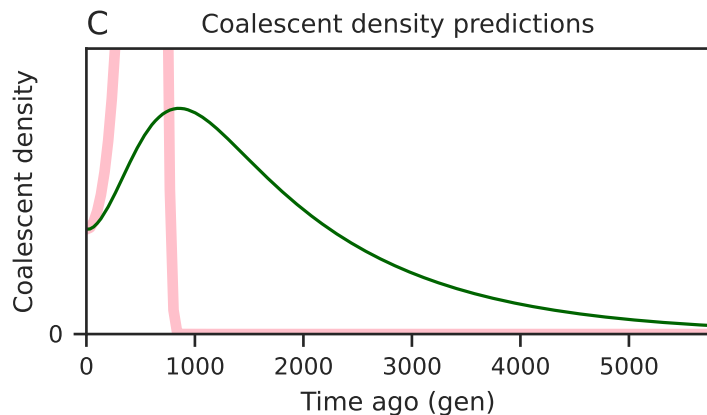
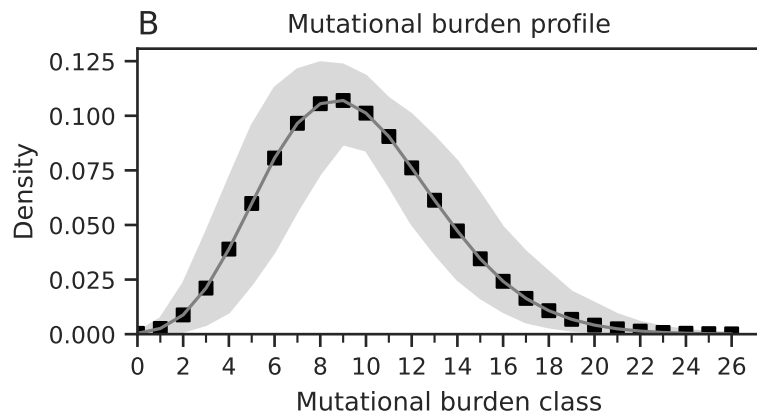
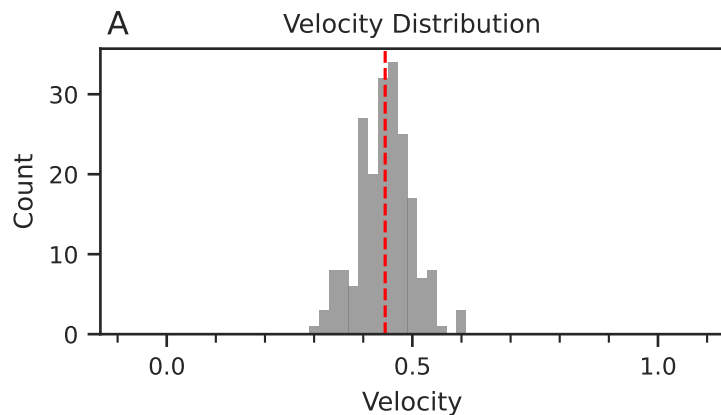
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 4 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4622 ± 0.0622 , $n = 200$ simulations



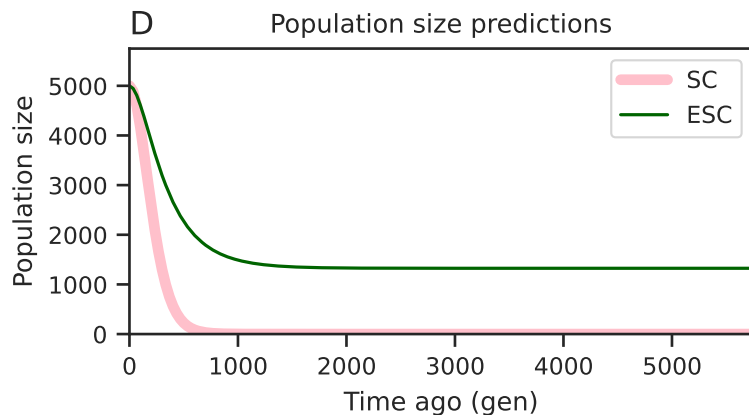
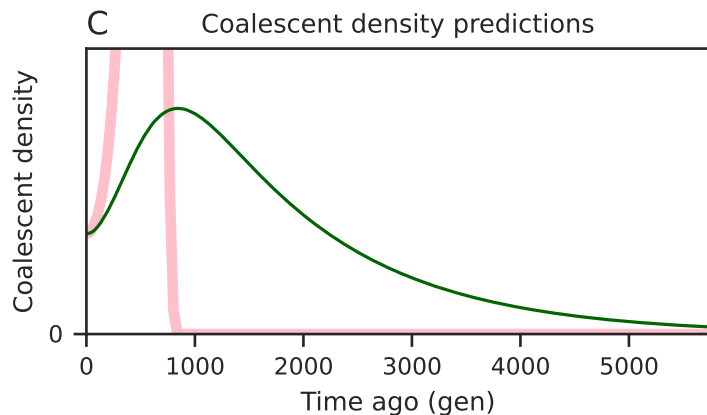
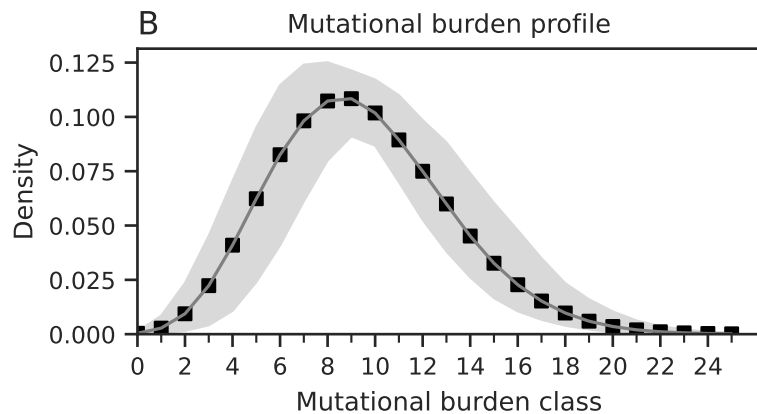
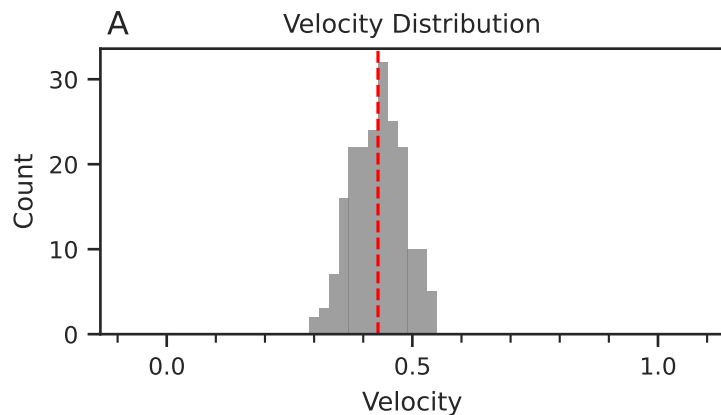
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4478 ± 0.0596 , $n = 200$ simulations



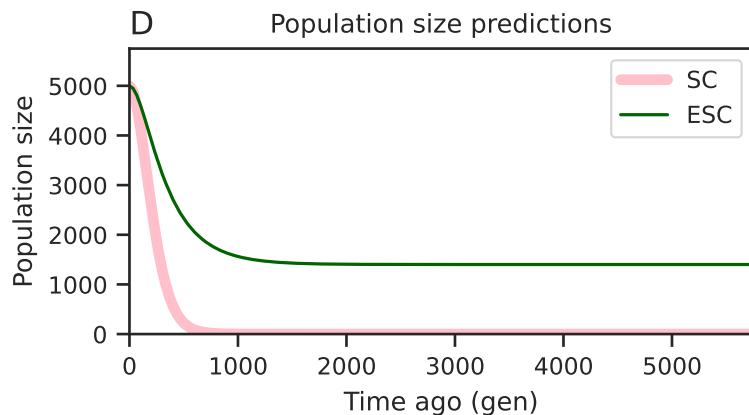
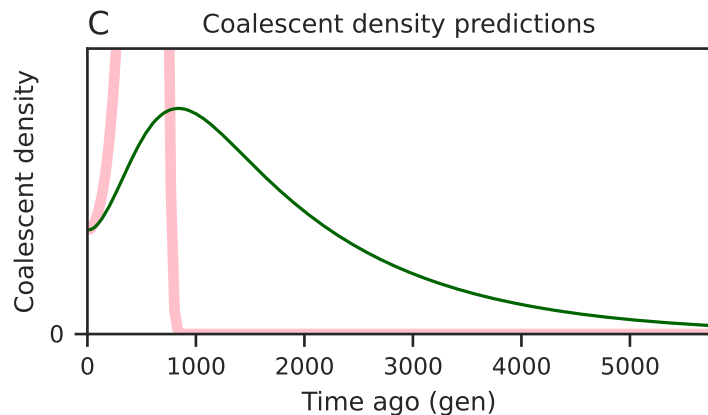
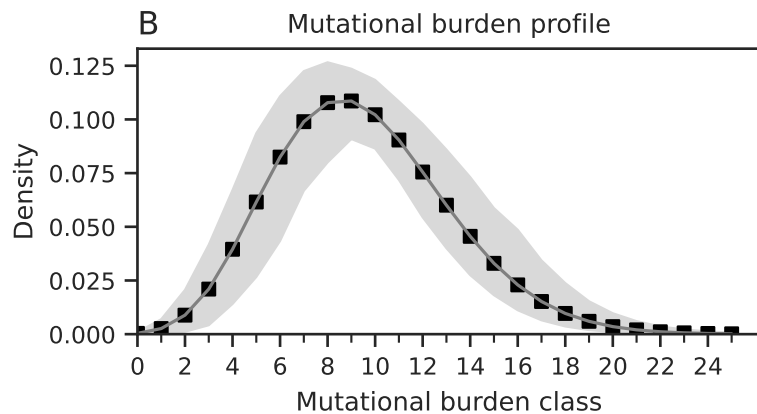
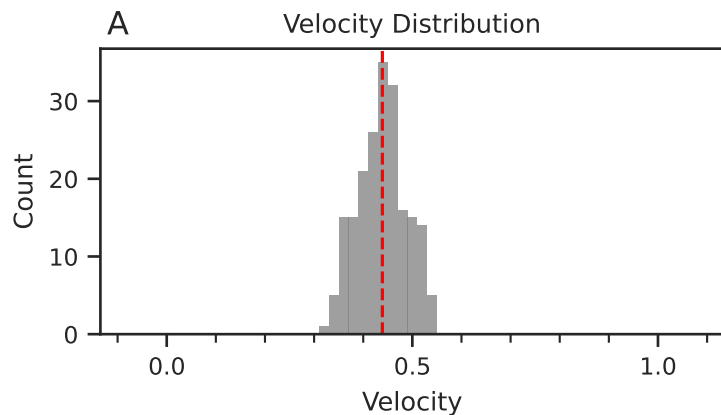
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4447 ± 0.0543 , $n = 200$ simulations



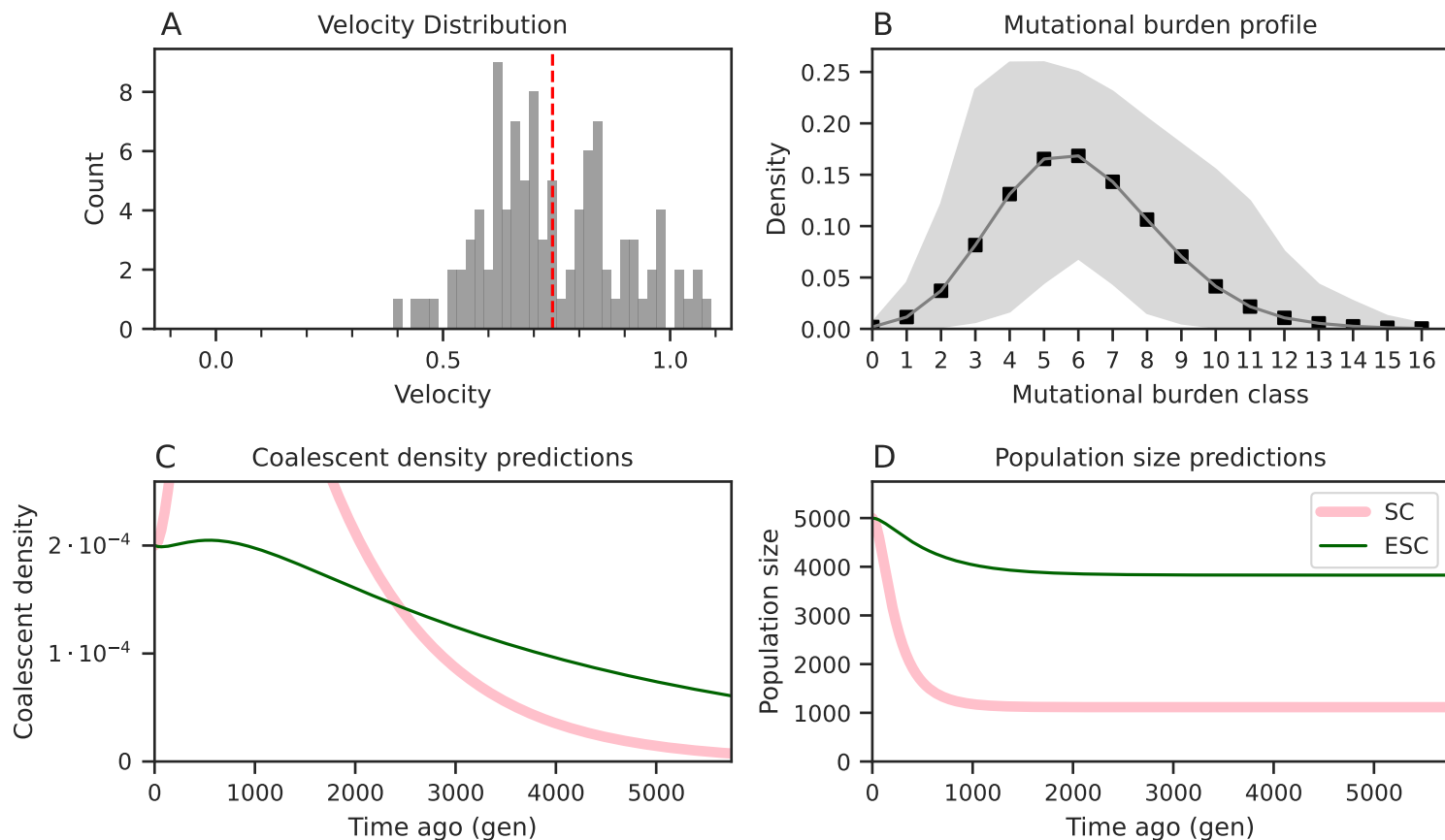
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4301 ± 0.0518 , $n = 200$ simulations



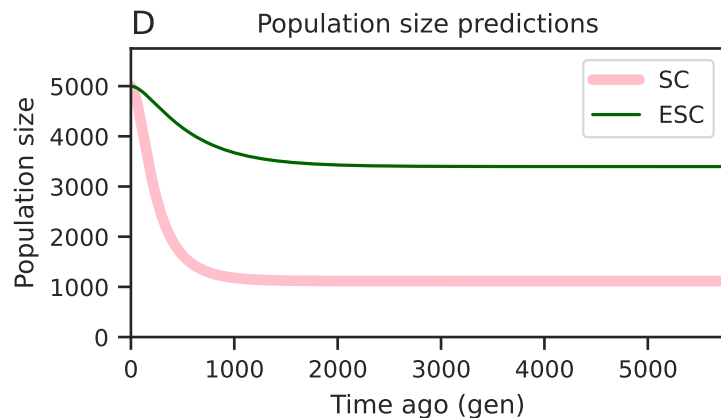
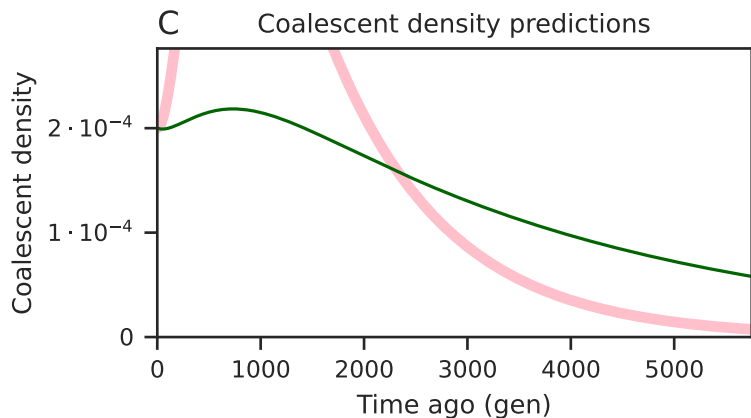
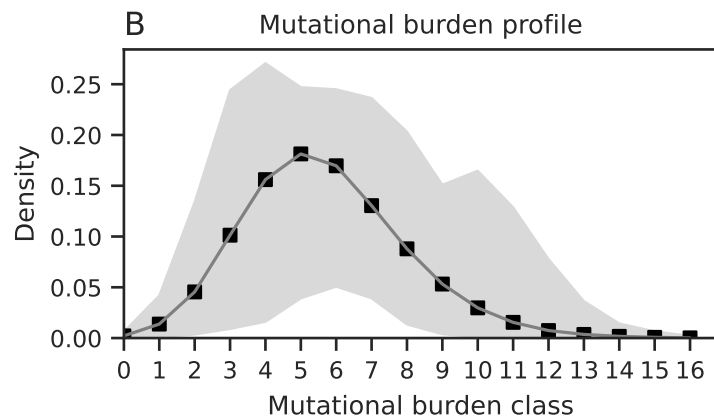
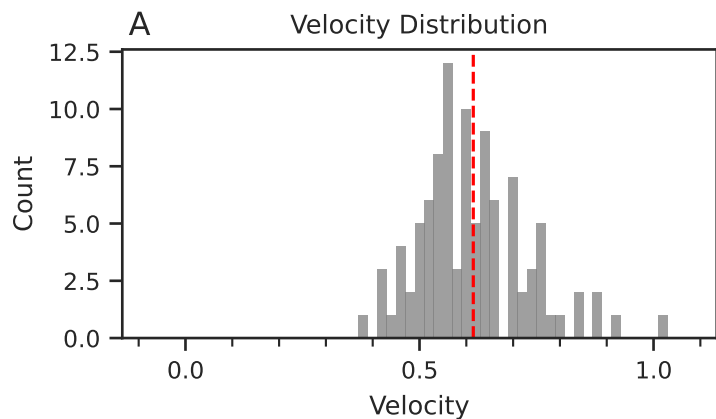
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\sigma = 0$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4389 ± 0.0494 , $n = 200$ simulations



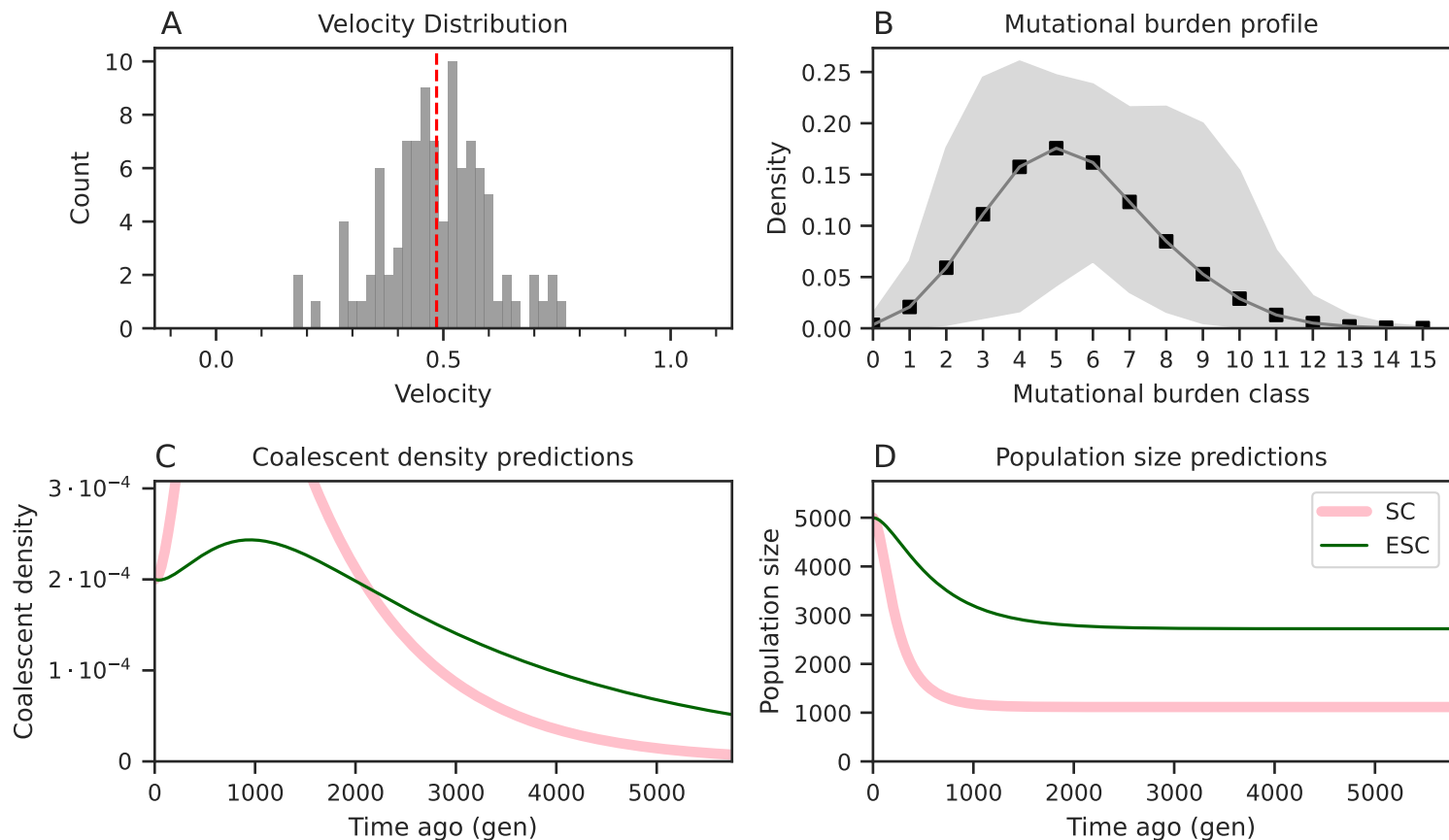
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.004$, $\phi = 4.46$
Mean Velocity = 0.7411 ± 0.1522 , $n = 100$ simulations



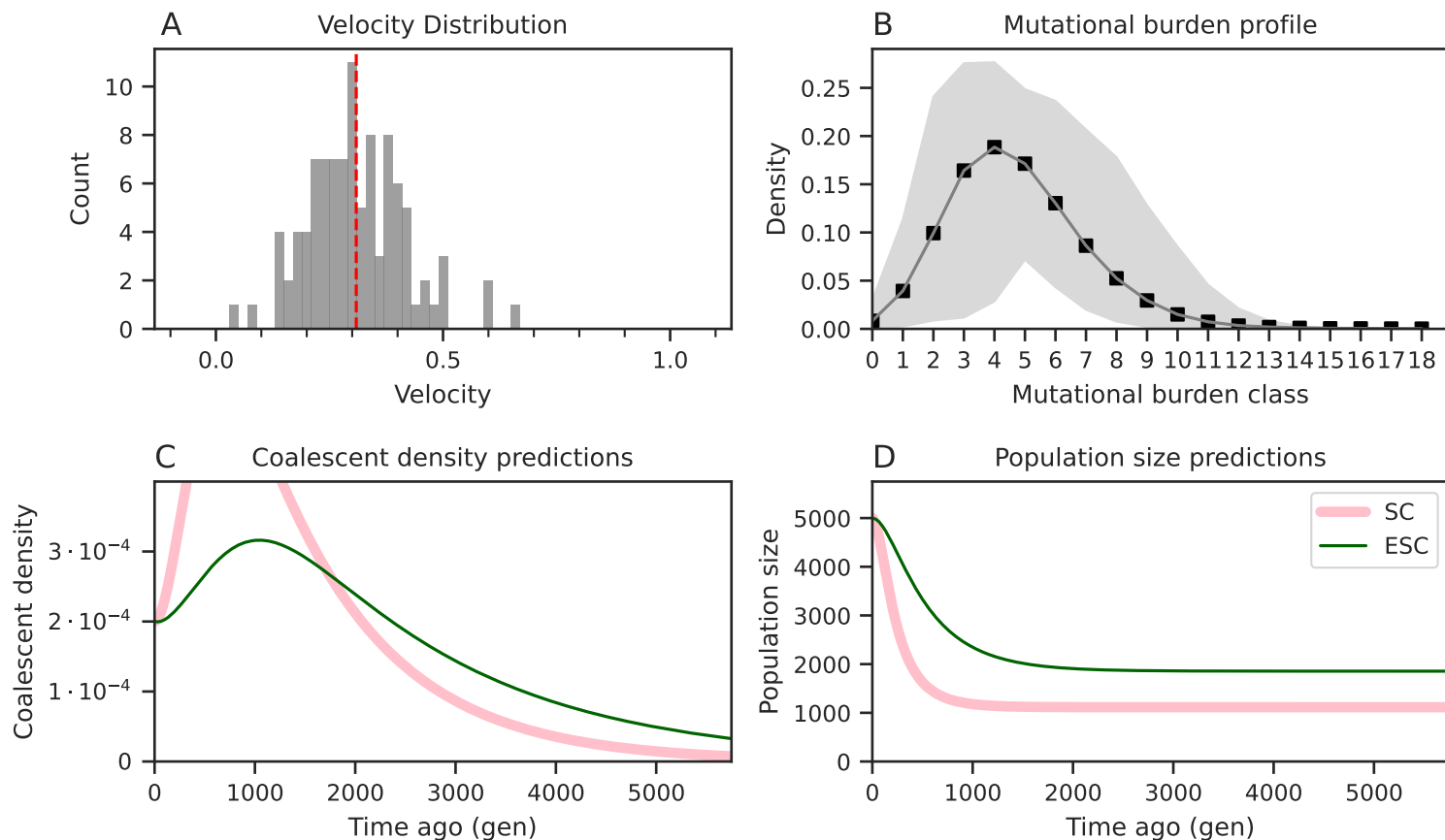
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.0036$, $\phi = 4.46$
Mean Velocity = 0.6149 ± 0.1155 , $n = 100$ simulations



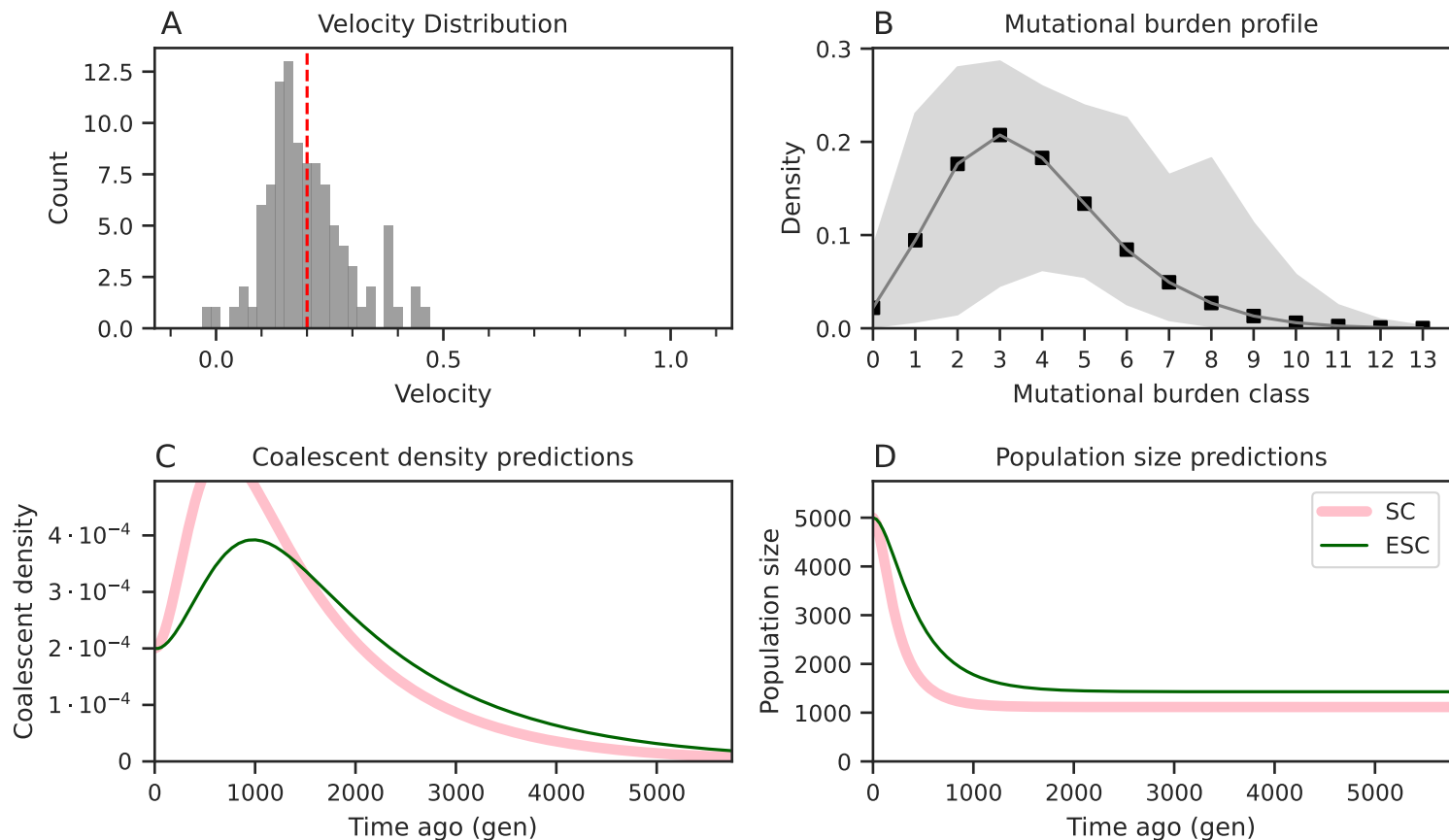
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.0032$, $\phi = 4.46$
Mean Velocity= 0.4851 ± 0.1157 , $n = 100$ simulations



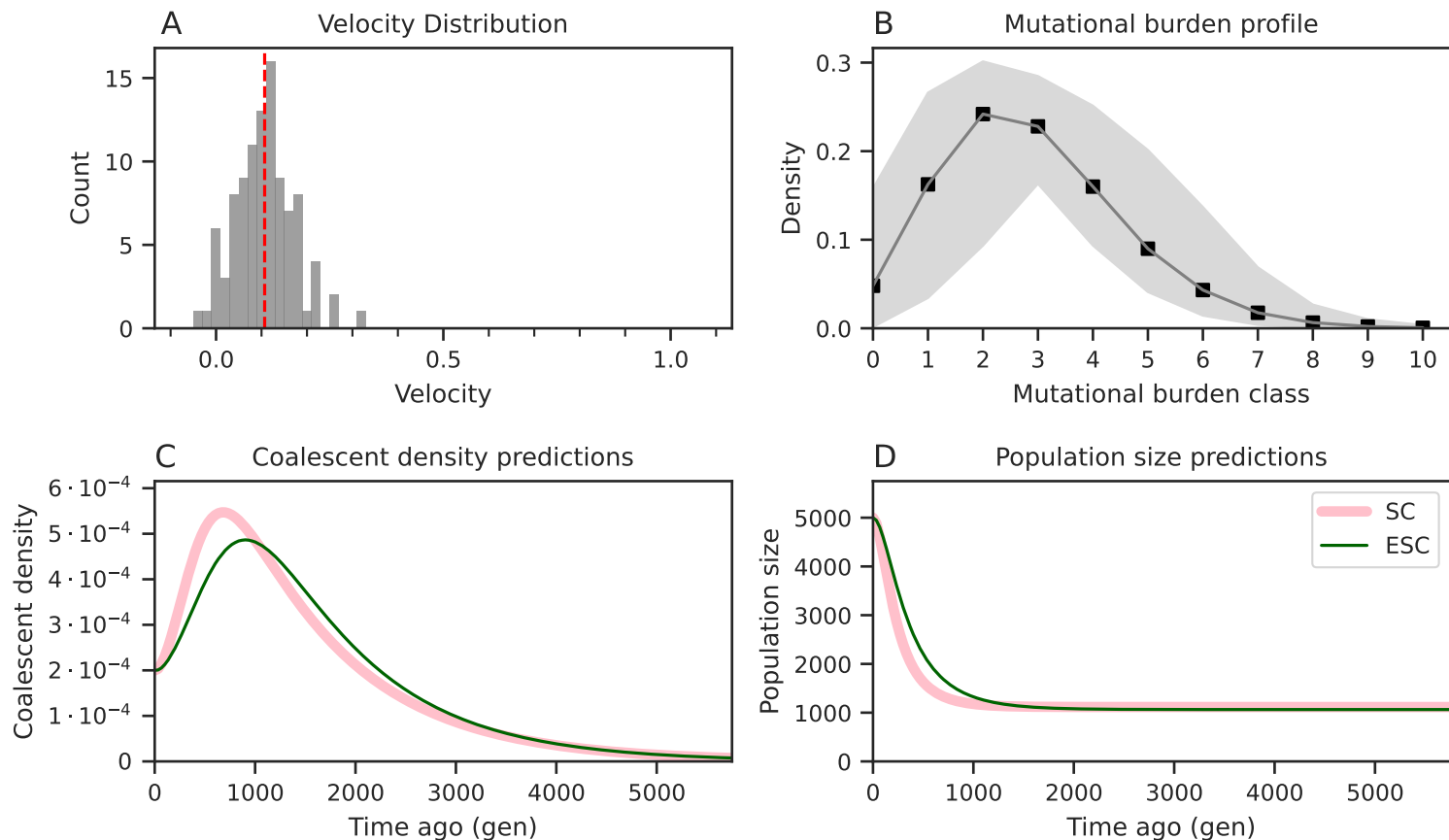
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.0028$, $\phi = 4.46$
Mean Velocity = 0.3090 ± 0.1074 , $n = 100$ simulations



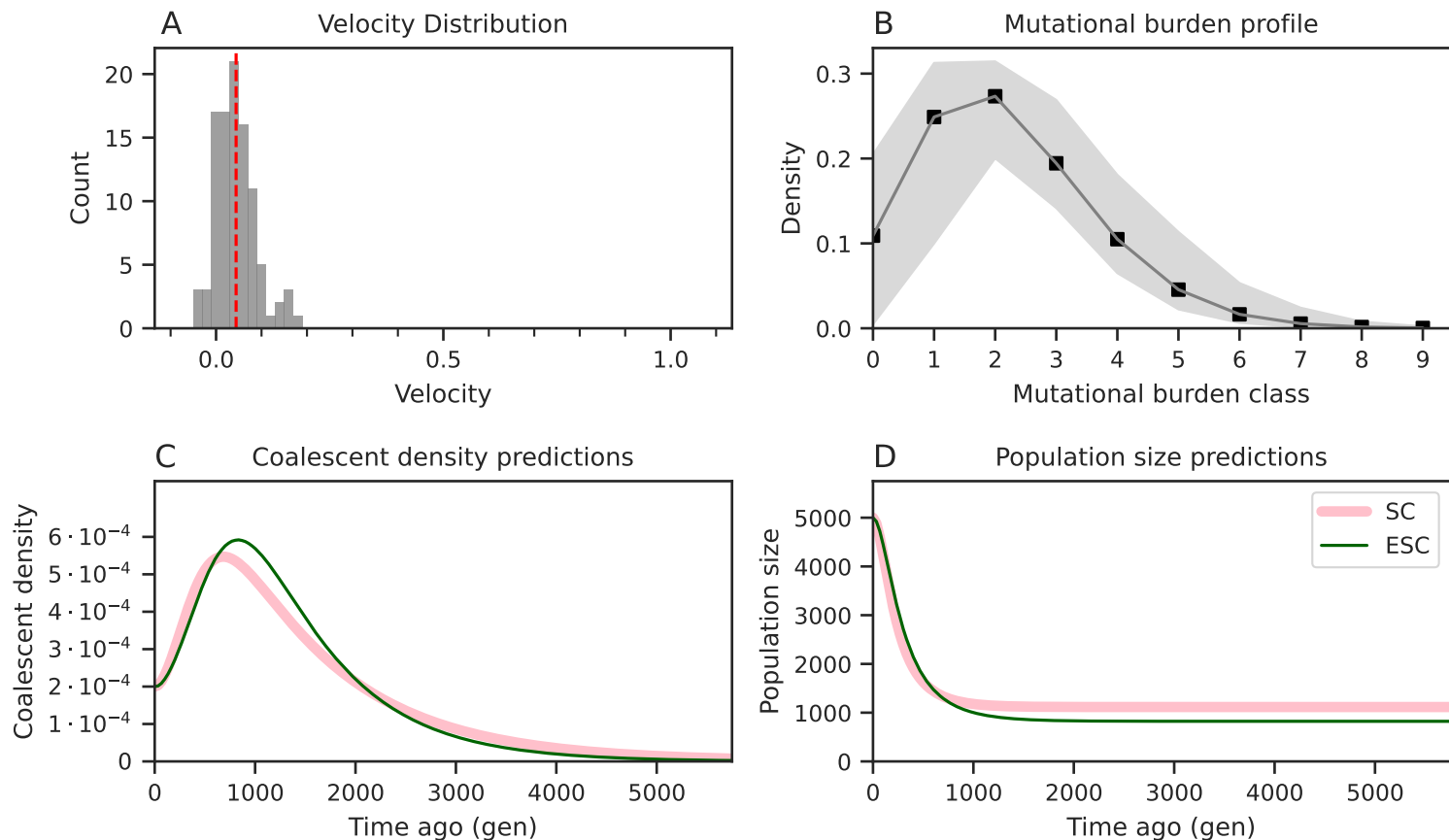
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.0024$, $\phi = 4.46$
Mean Velocity= 0.2003 ± 0.0931 , $n = 100$ simulations



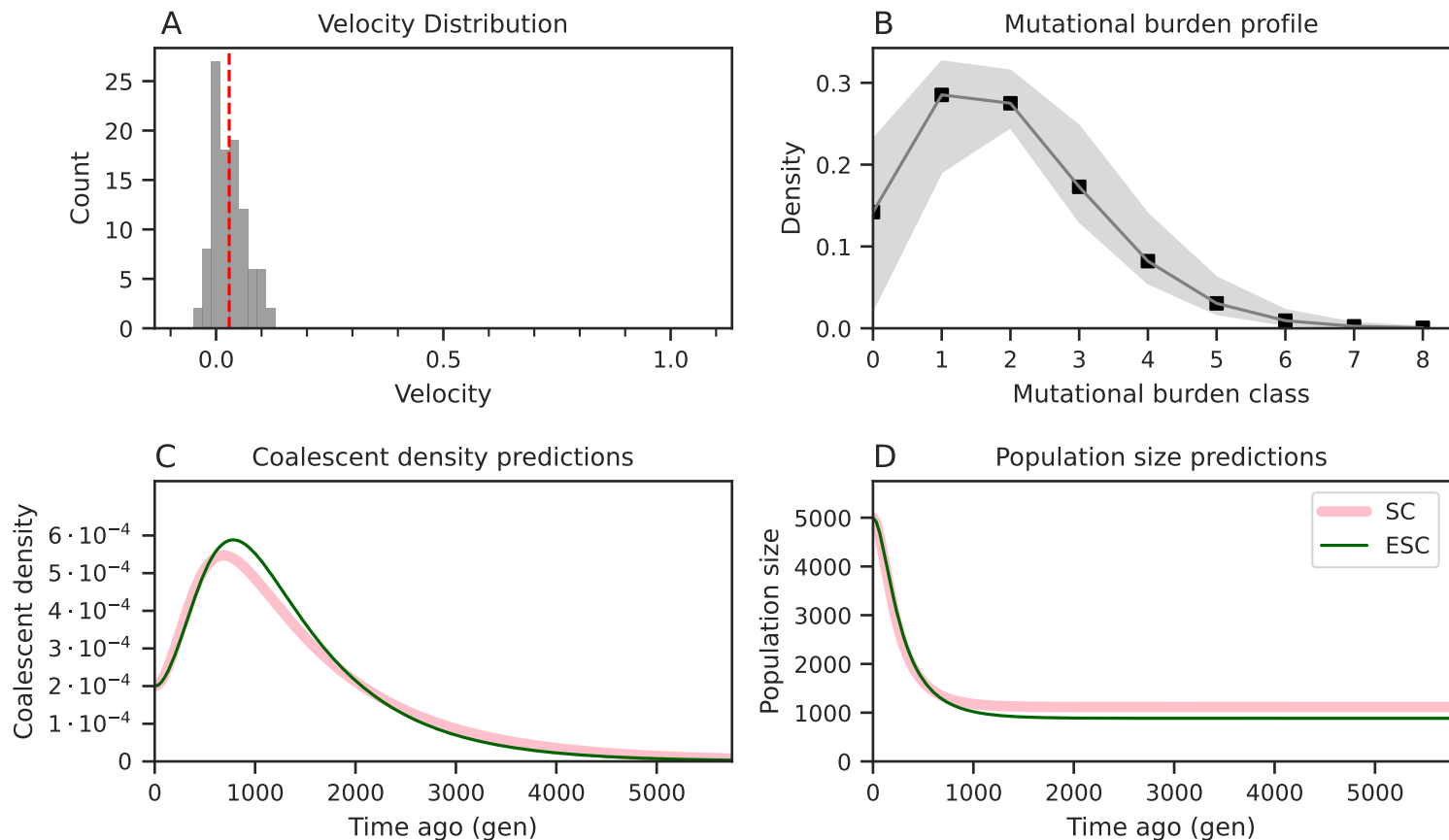
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.002$, $\phi = 4.46$
Mean Velocity= 0.1069 ± 0.0652 , $n = 100$ simulations



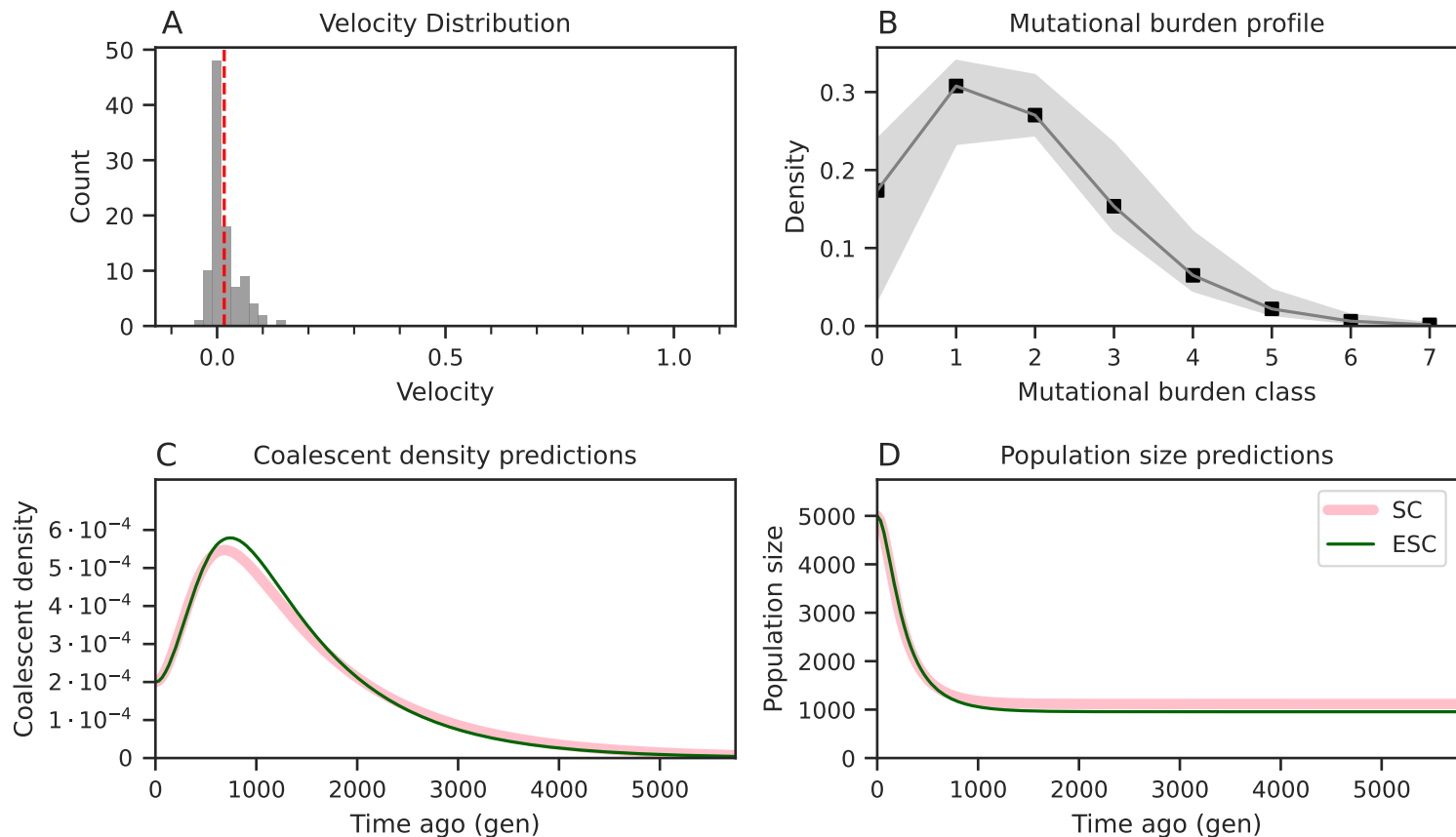
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.0016$, $\phi = 4.46$
Mean Velocity= 0.0441 ± 0.0439 , $n = 100$ simulations



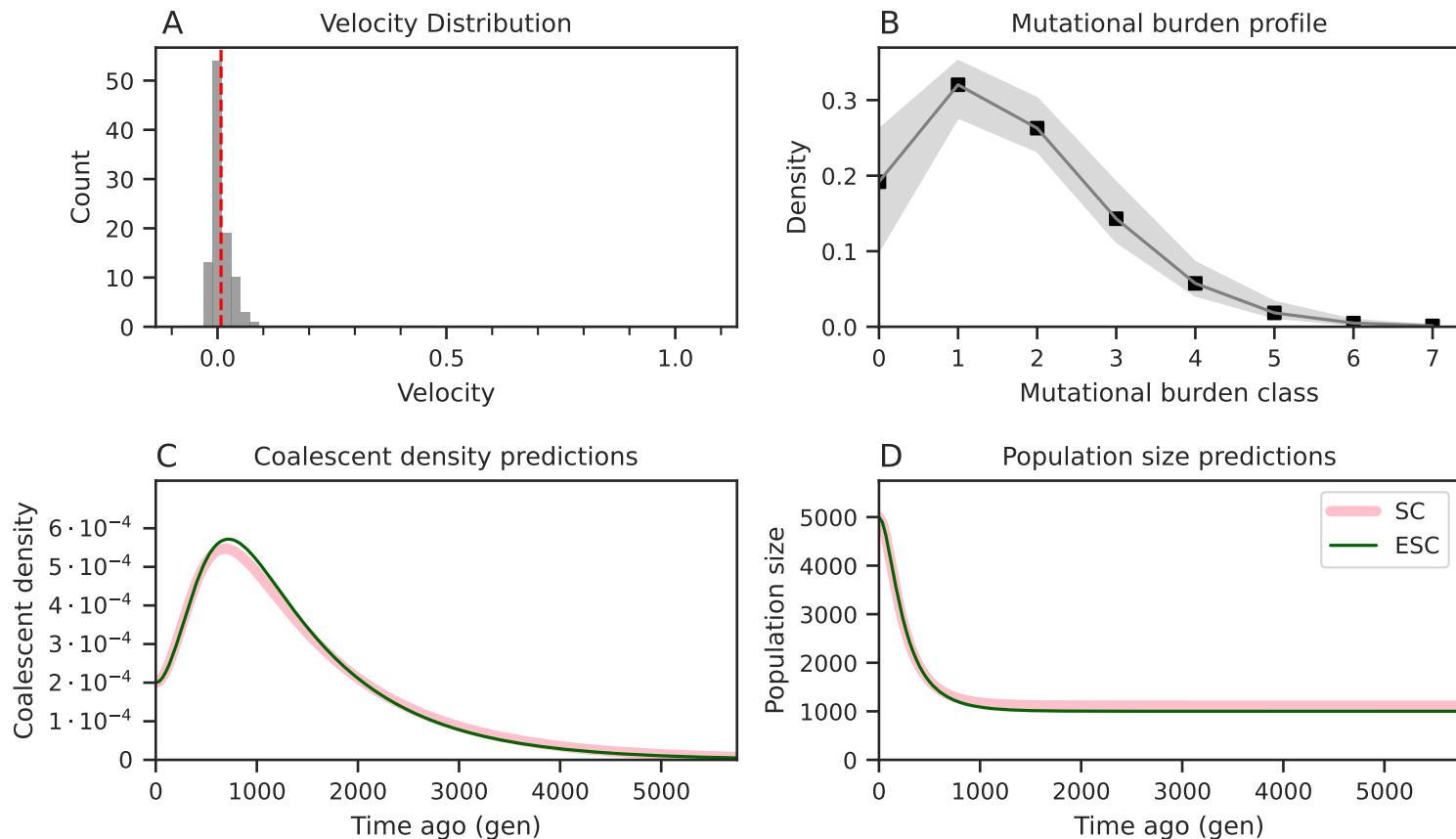
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.0012$, $\phi = 4.46$
Mean Velocity= 0.0288 ± 0.0352 , $n = 100$ simulations



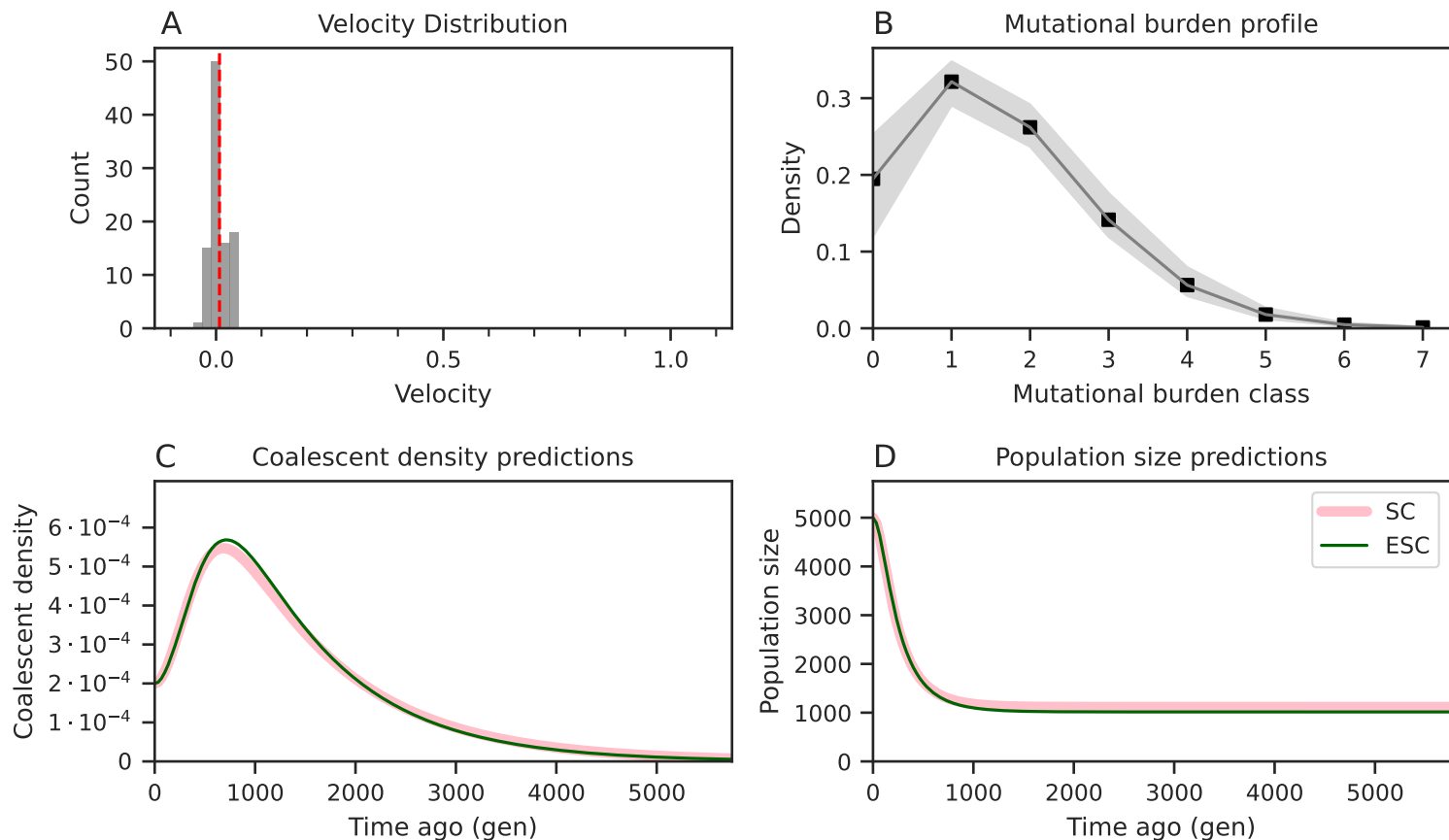
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 8 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0156 ± 0.0299 , $n = 100$ simulations



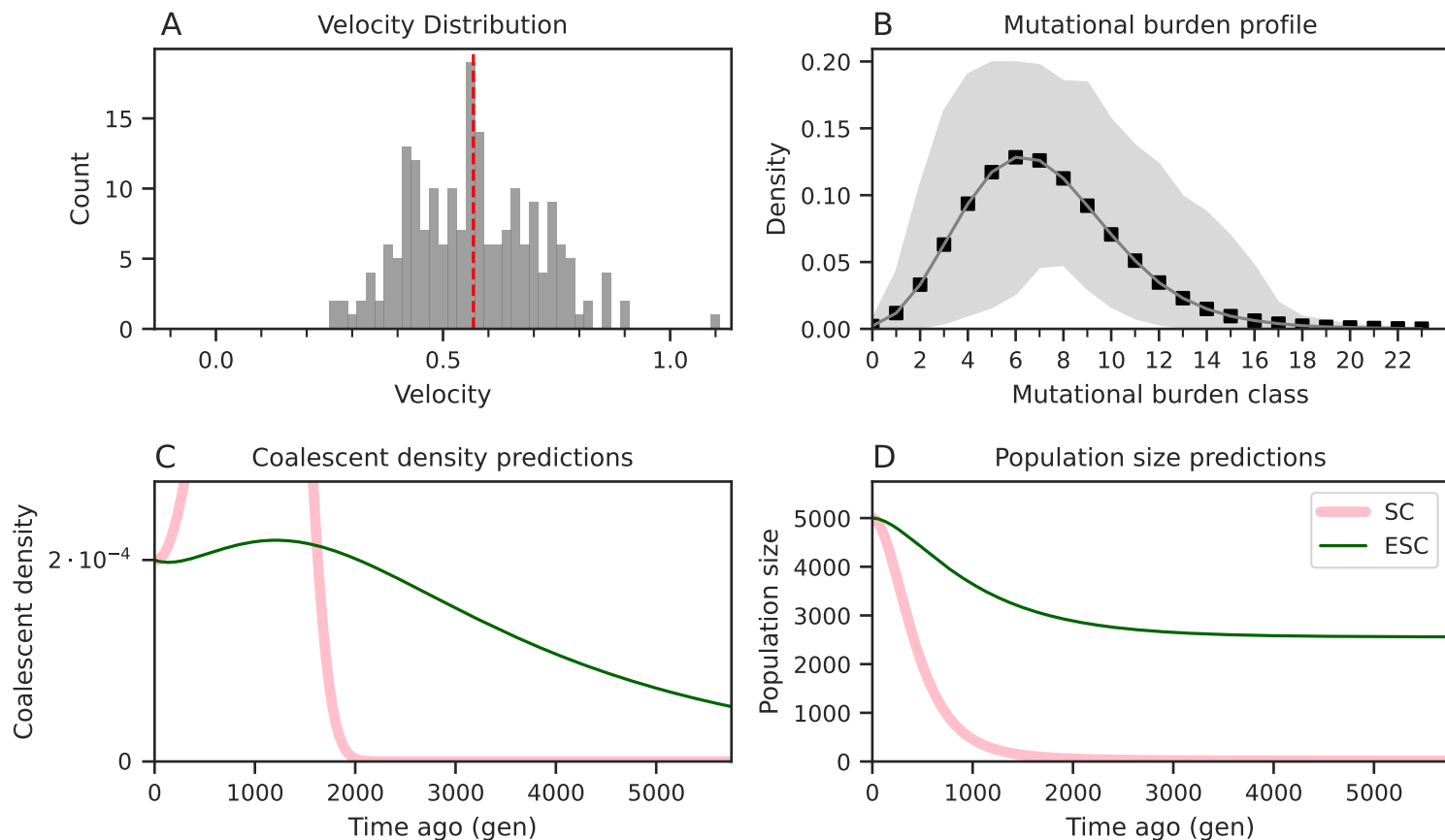
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 4 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0077 ± 0.0188 , $n = 100$ simulations



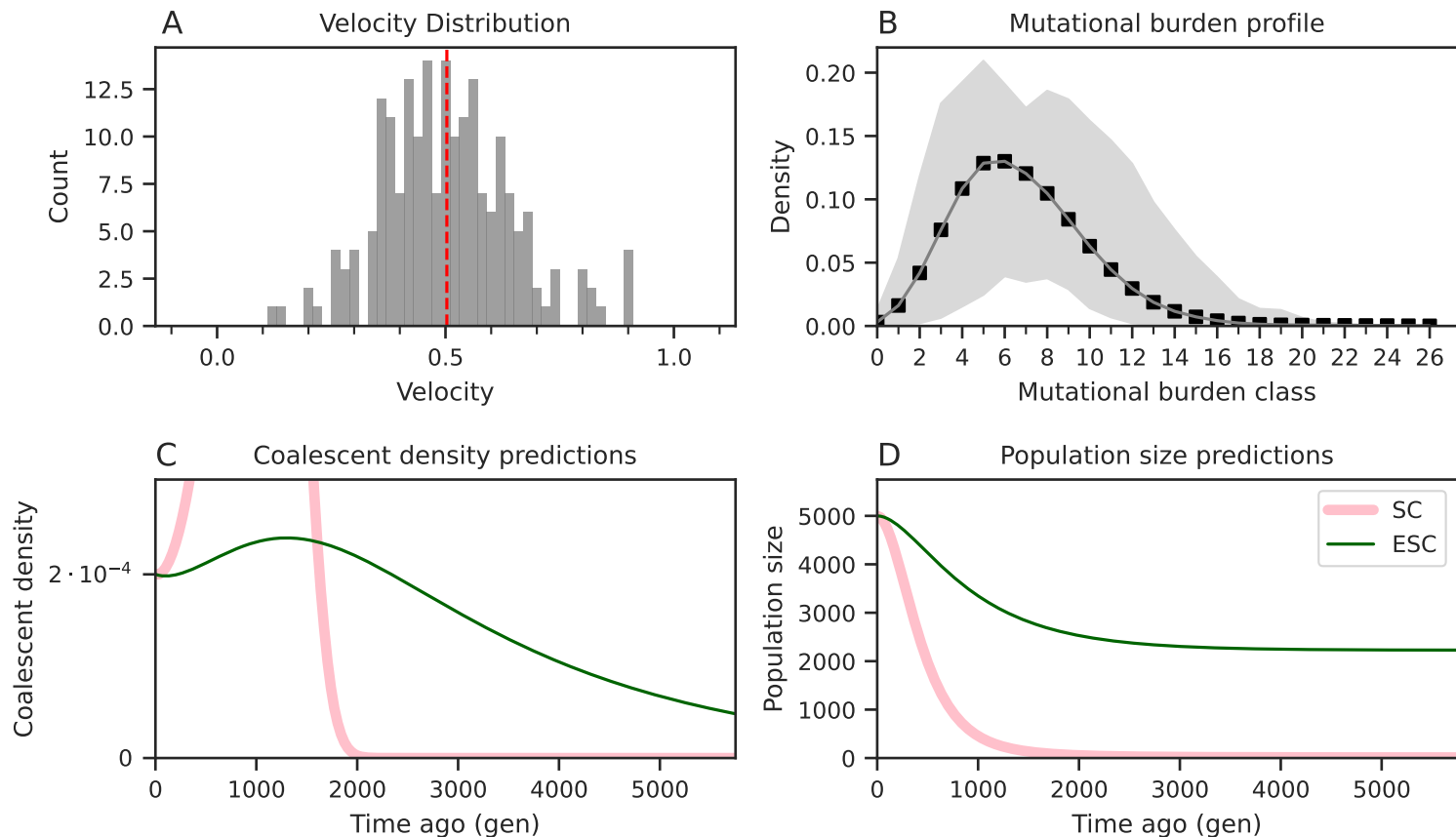
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0$, $\phi = 4.46$
Mean Velocity= 0.0076 ± 0.0194 , $n = 100$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 0.0124$
Mean Velocity = 0.5667 ± 0.1430 , $n = 200$ simulations

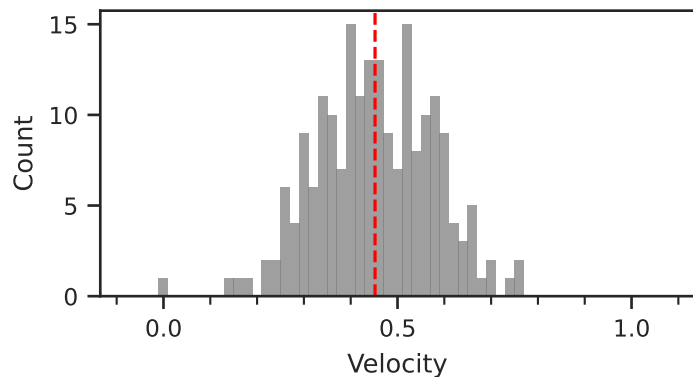


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 9 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.5030 ± 0.1423 , $n = 200$ simulations

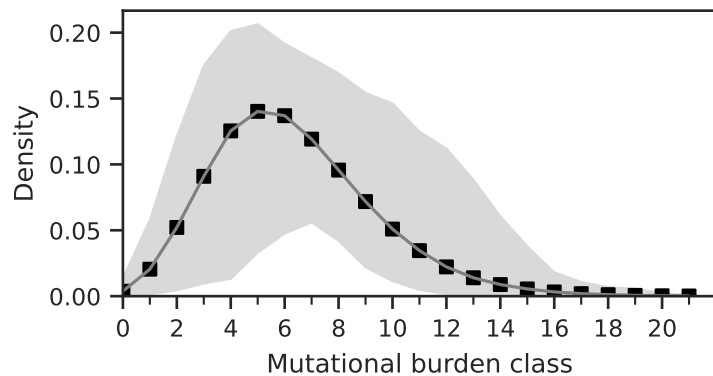


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 8 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.4520 ± 0.1253 , $n = 200$ simulations

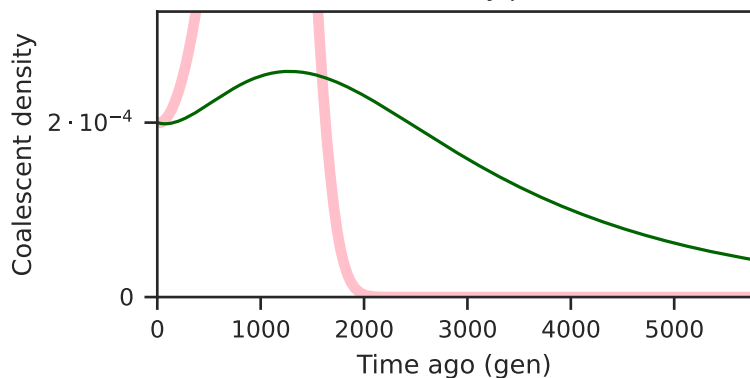
A Velocity Distribution



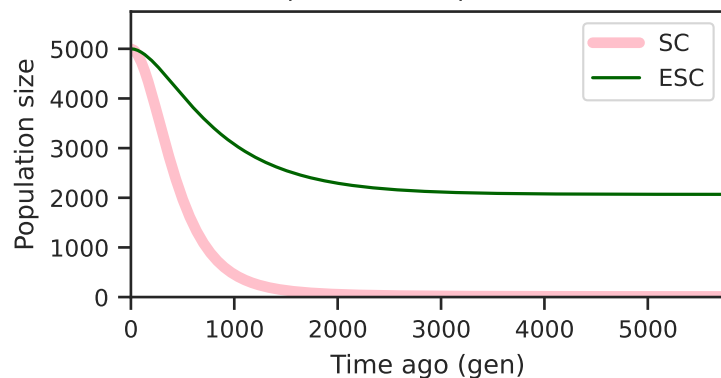
B Mutational burden profile



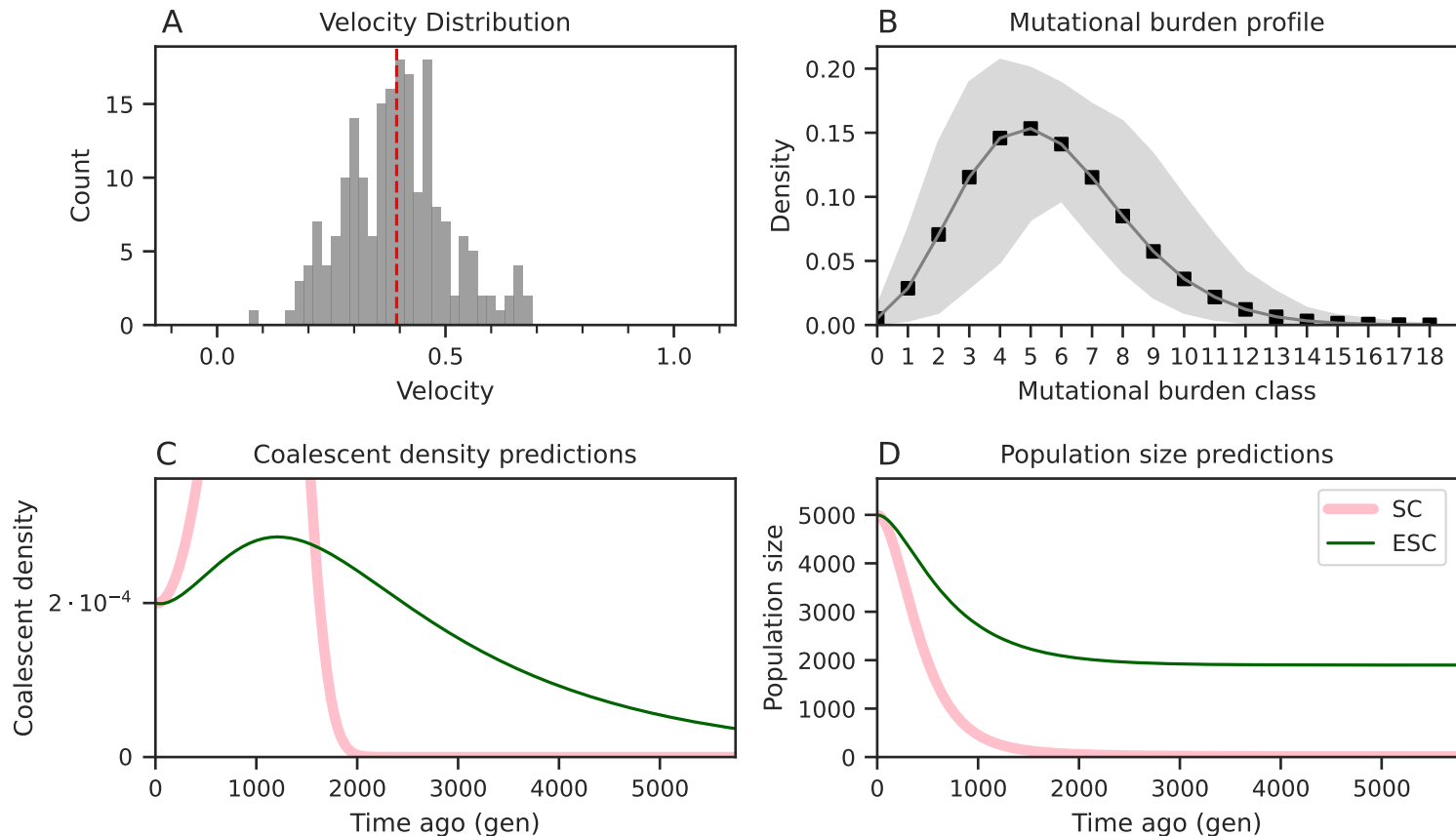
C Coalescent density predictions



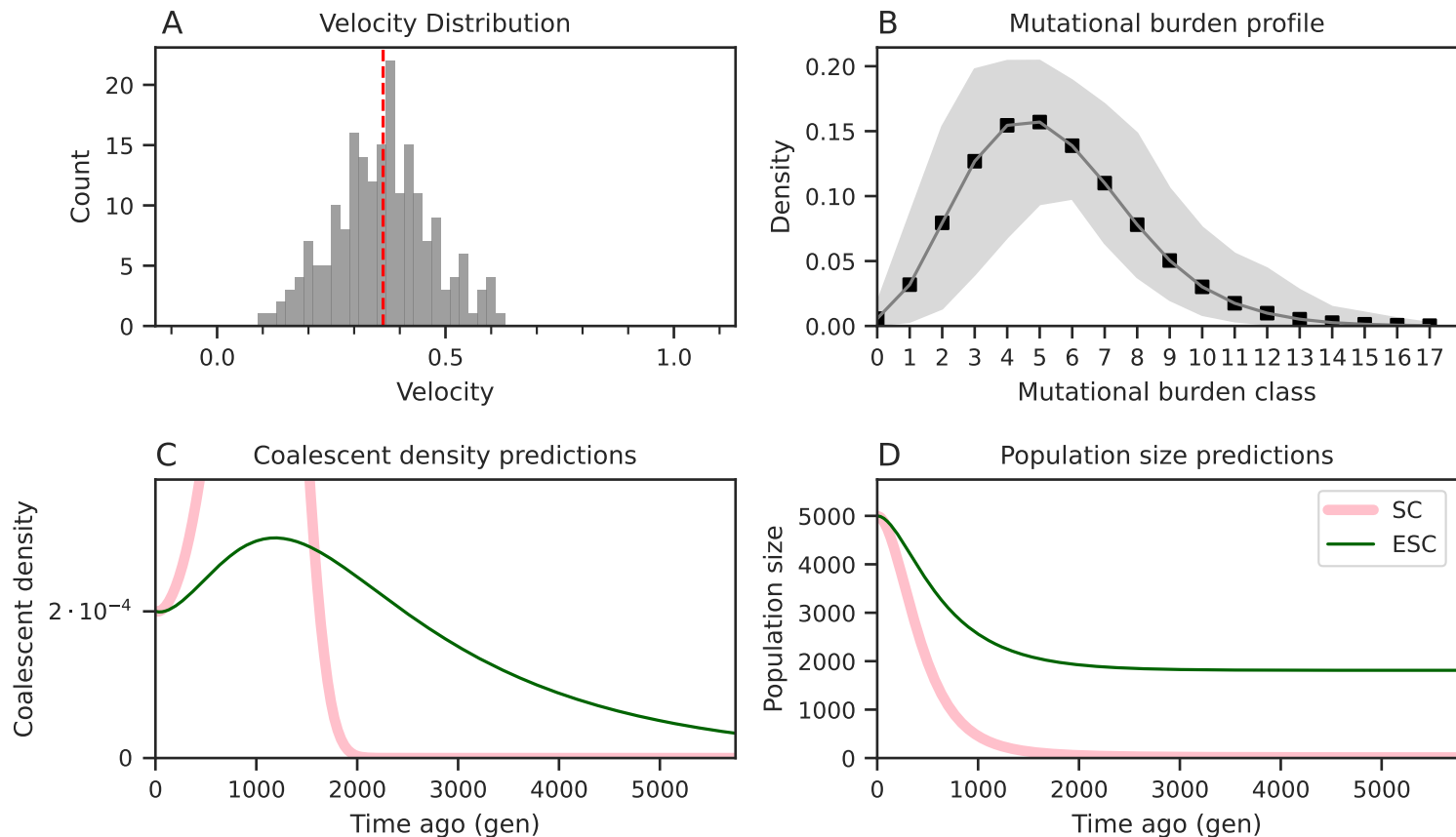
D Population size predictions



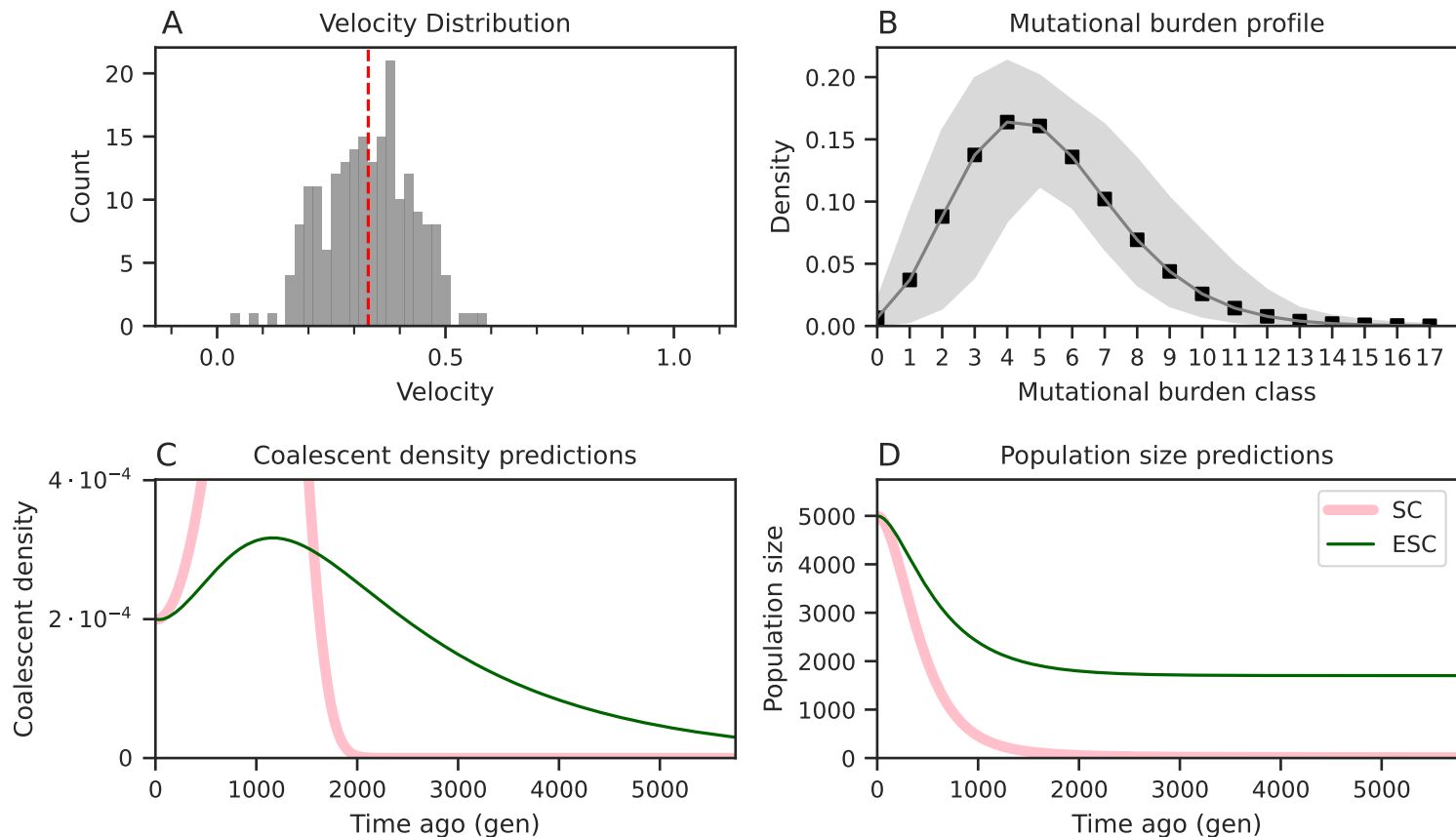
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 7 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3930 ± 0.1120 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 6 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3635 ± 0.1054 , $n = 200$ simulations

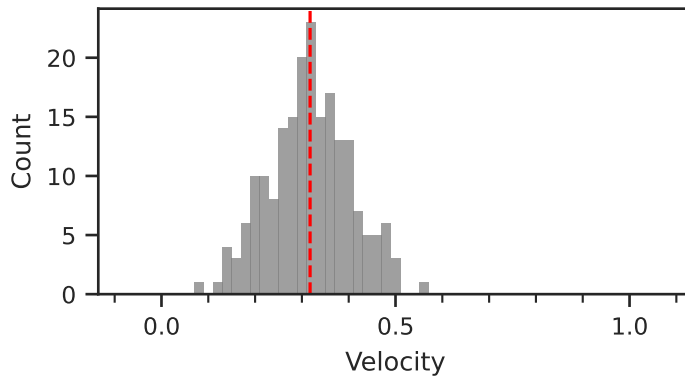


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3309 ± 0.0965 , $n = 200$ simulations

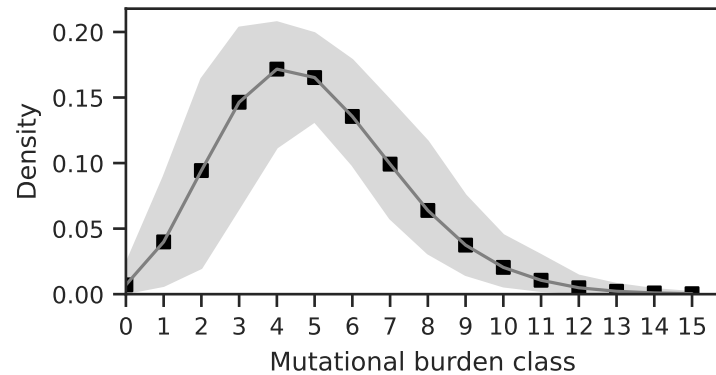


$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 4 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.3175 ± 0.0863 , $n = 200$ simulations

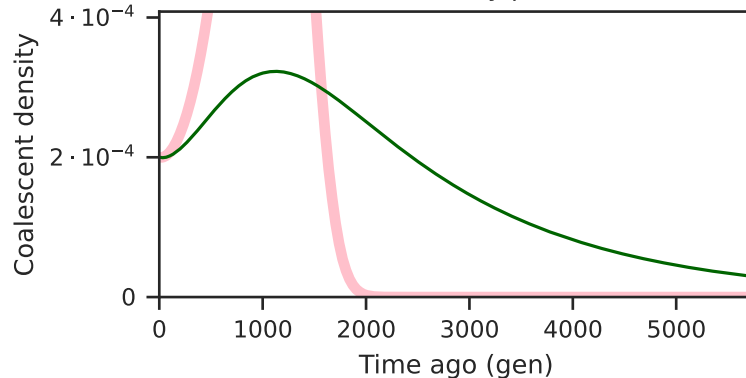
A Velocity Distribution



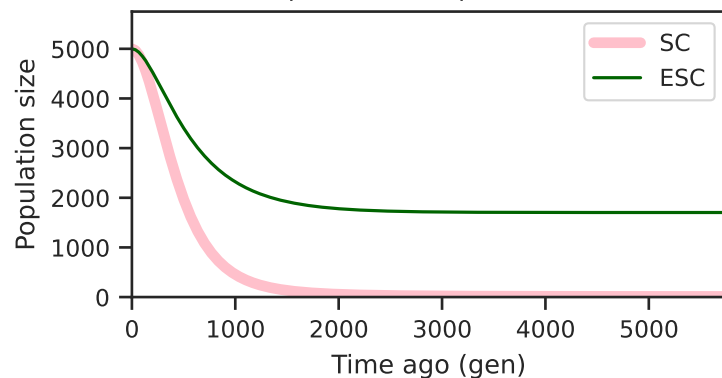
B Mutational burden profile



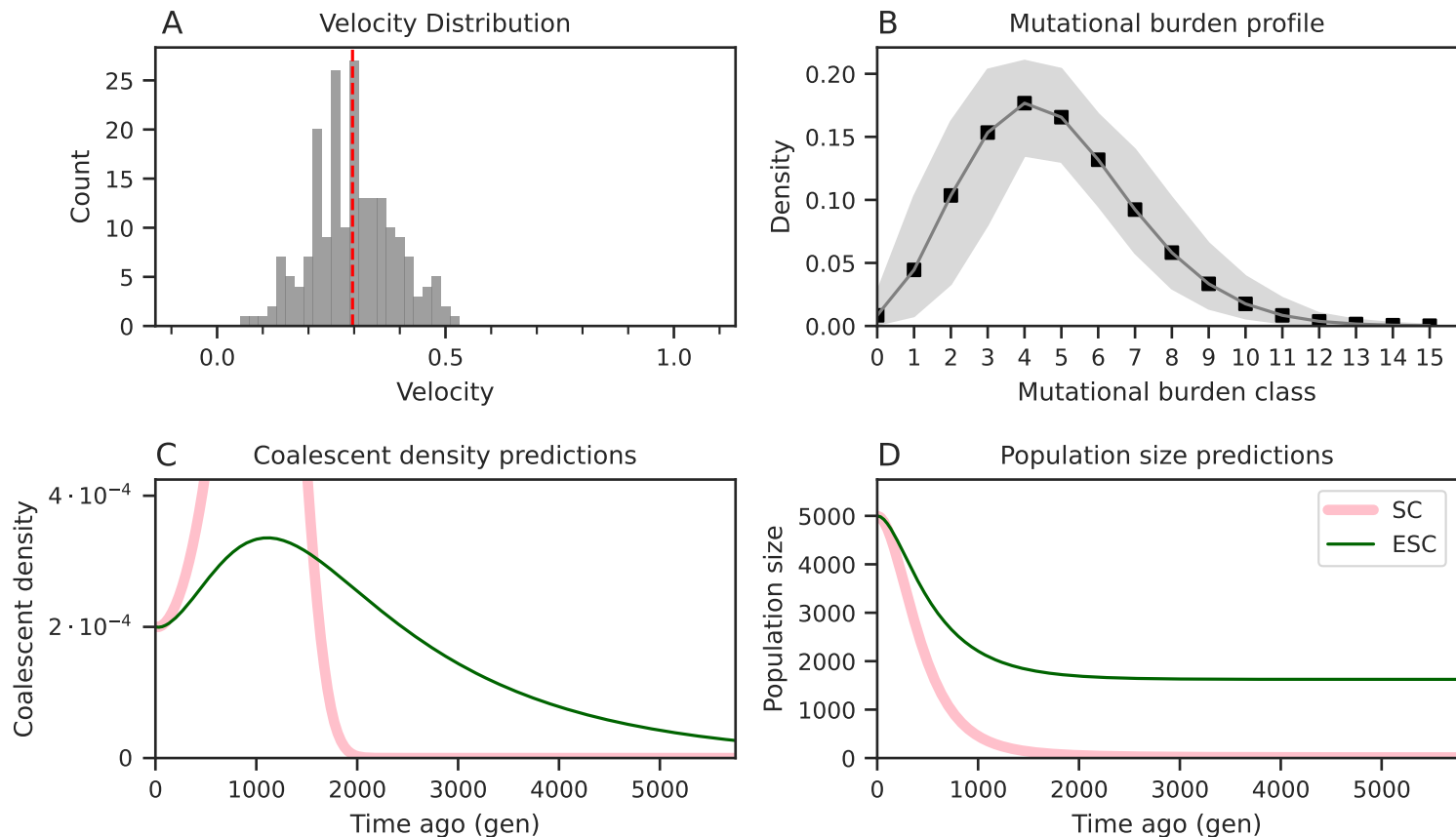
C Coalescent density predictions



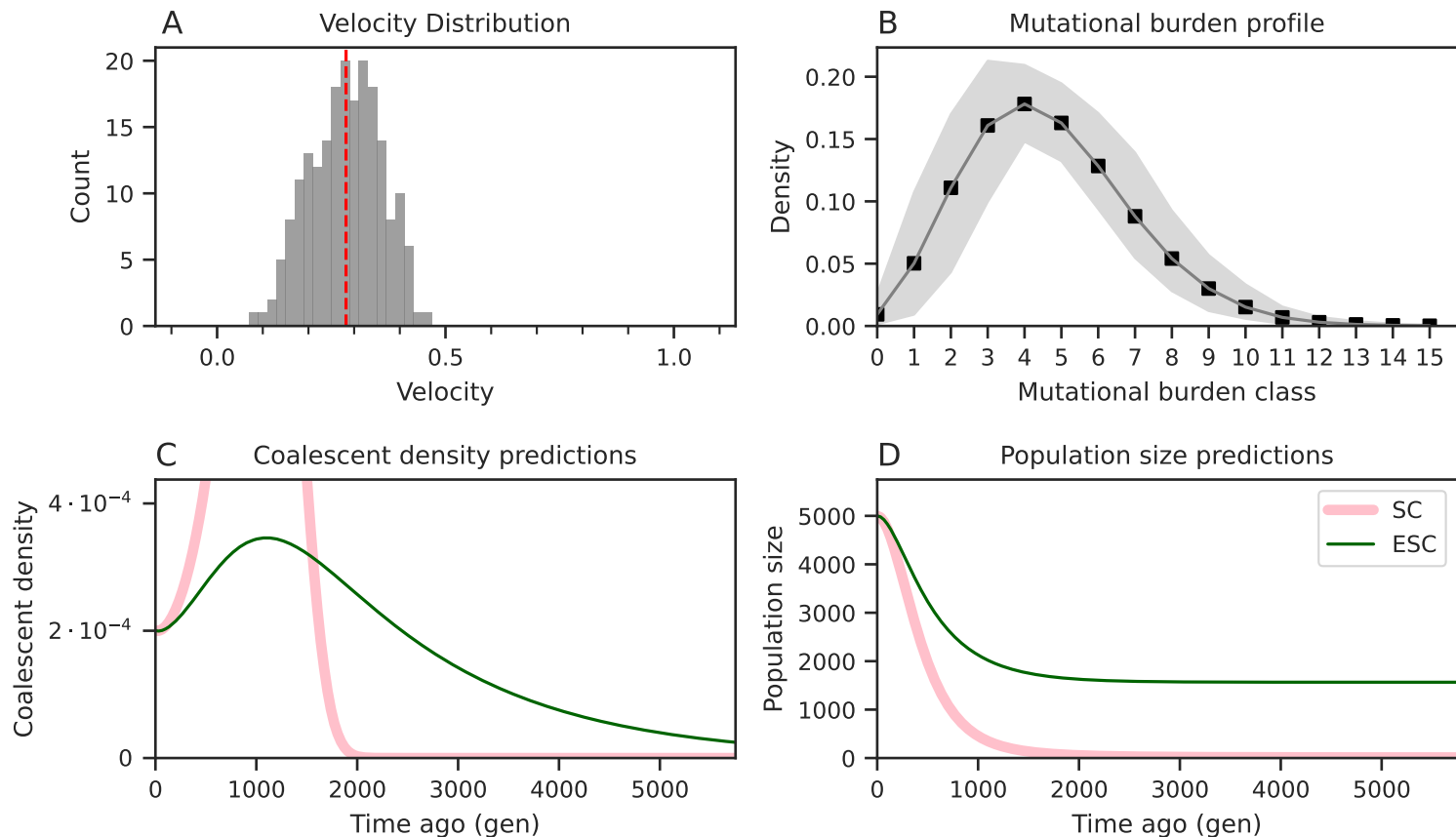
D Population size predictions



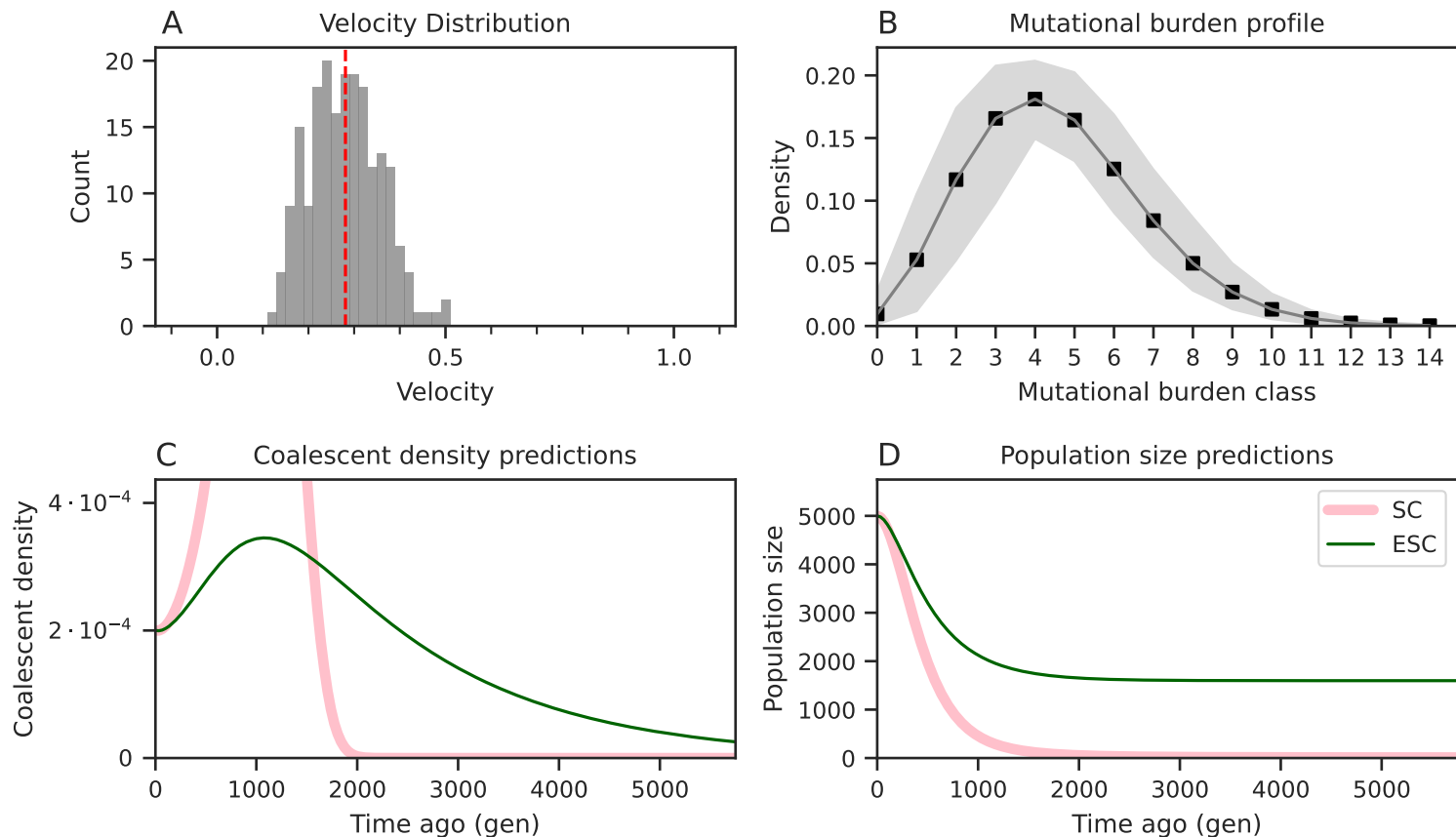
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.2965 ± 0.0886 , $n = 200$ simulations



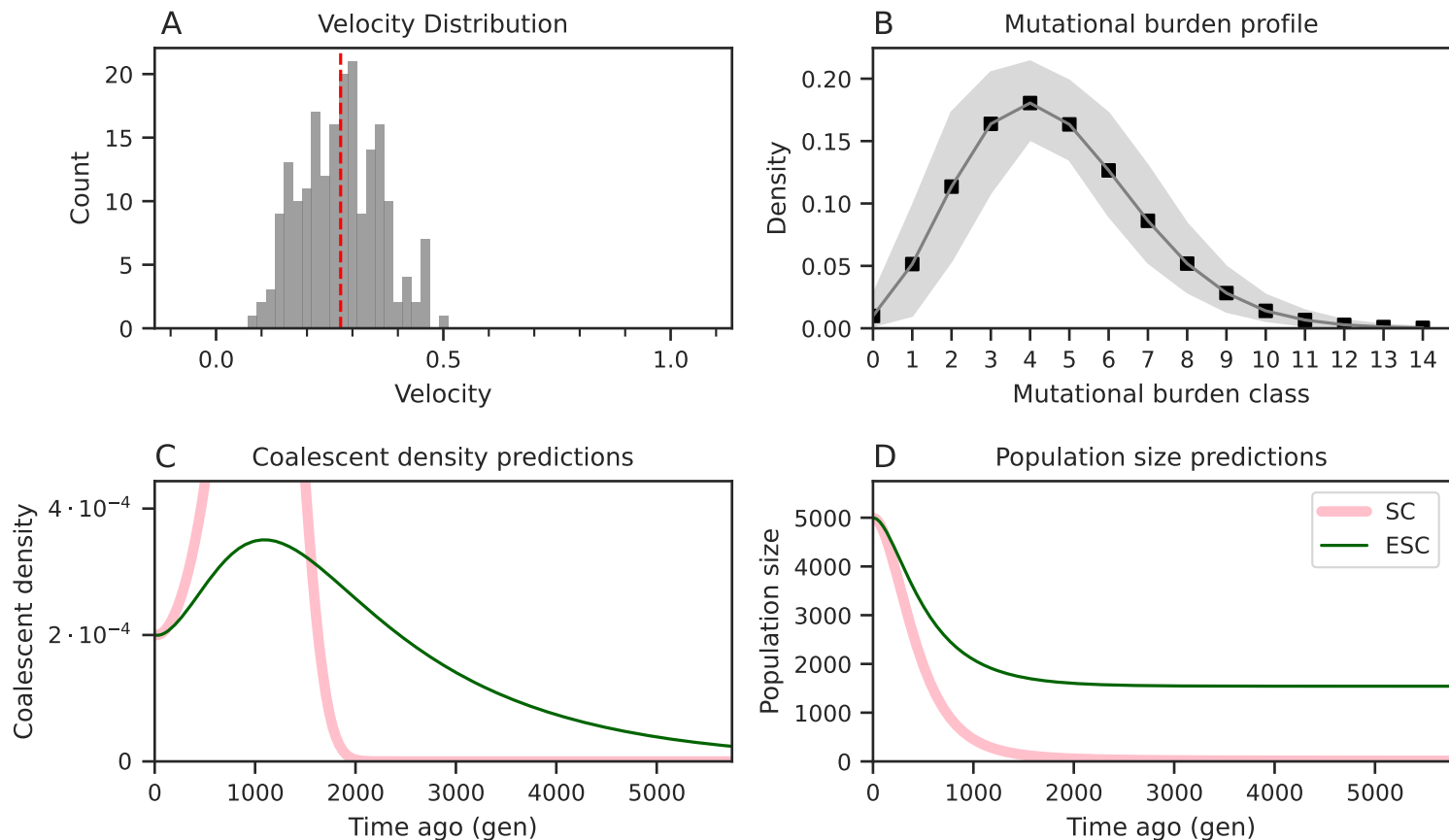
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity = 0.2823 ± 0.0768 , $n = 200$ simulations



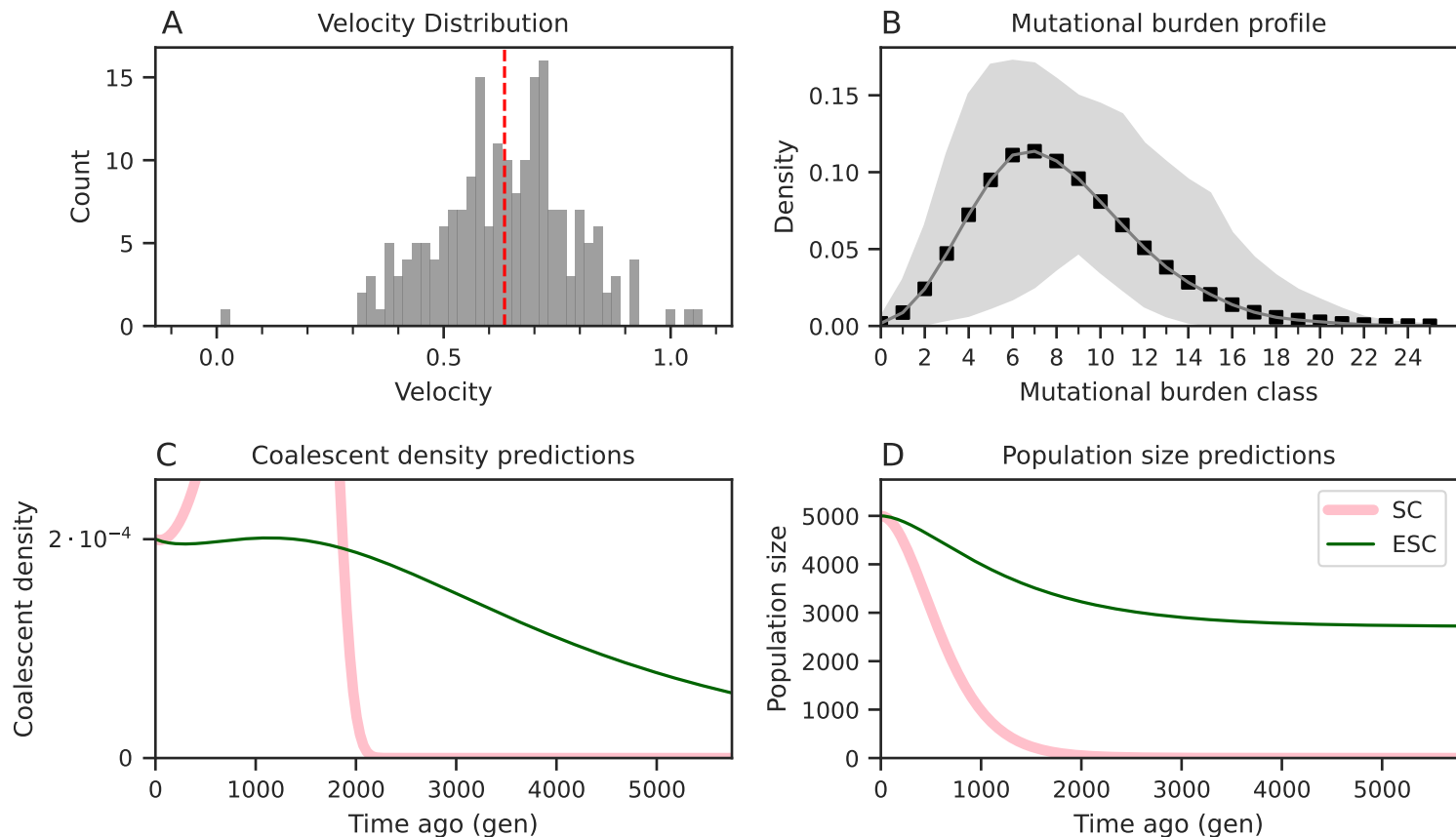
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 0.0124$
Mean Velocity= 0.2810 ± 0.0762 , $n = 200$ simulations



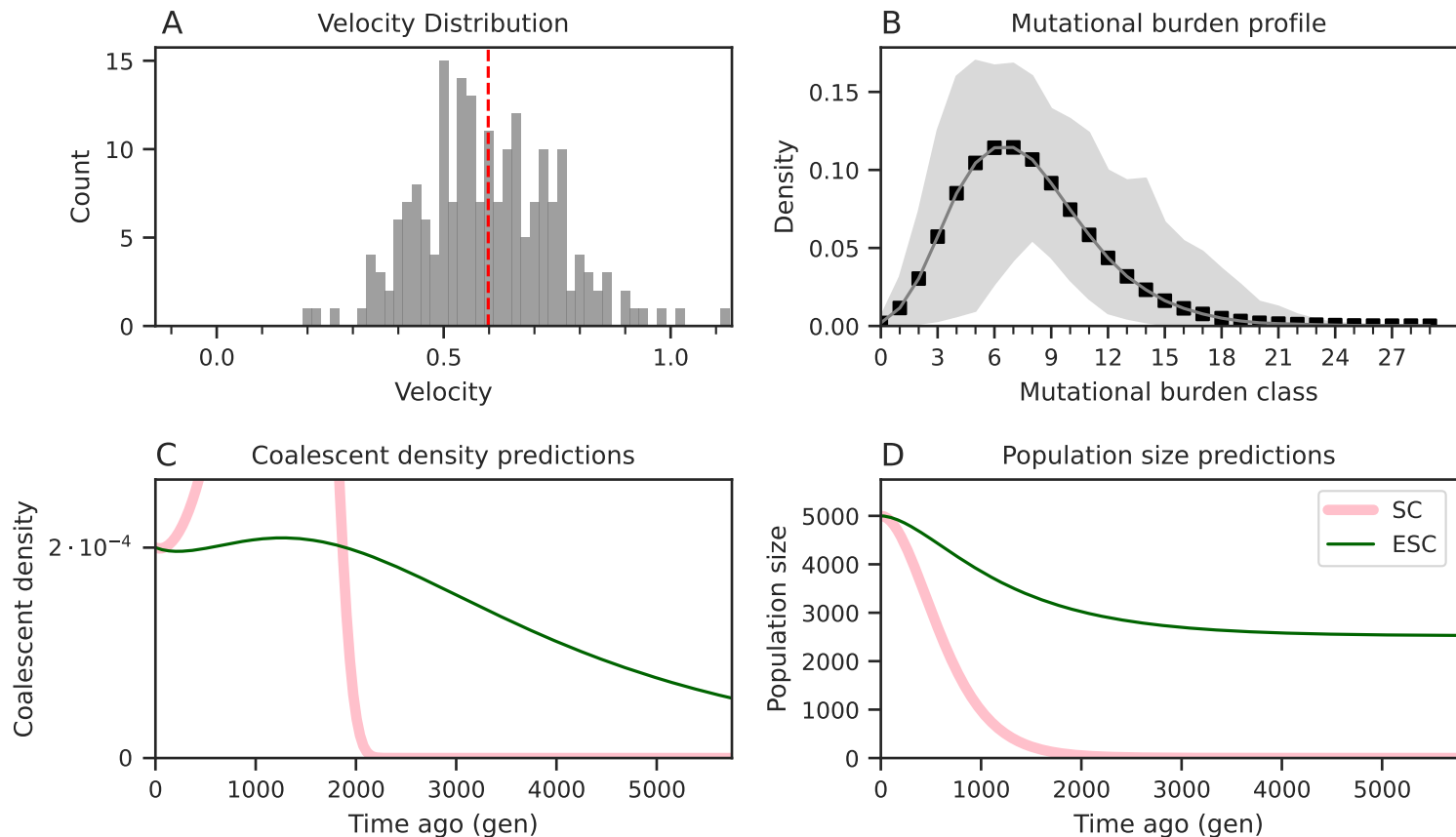
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\sigma = 0$, $\phi = 0.0124$
Mean Velocity = 0.2740 ± 0.0860 , $n = 200$ simulations



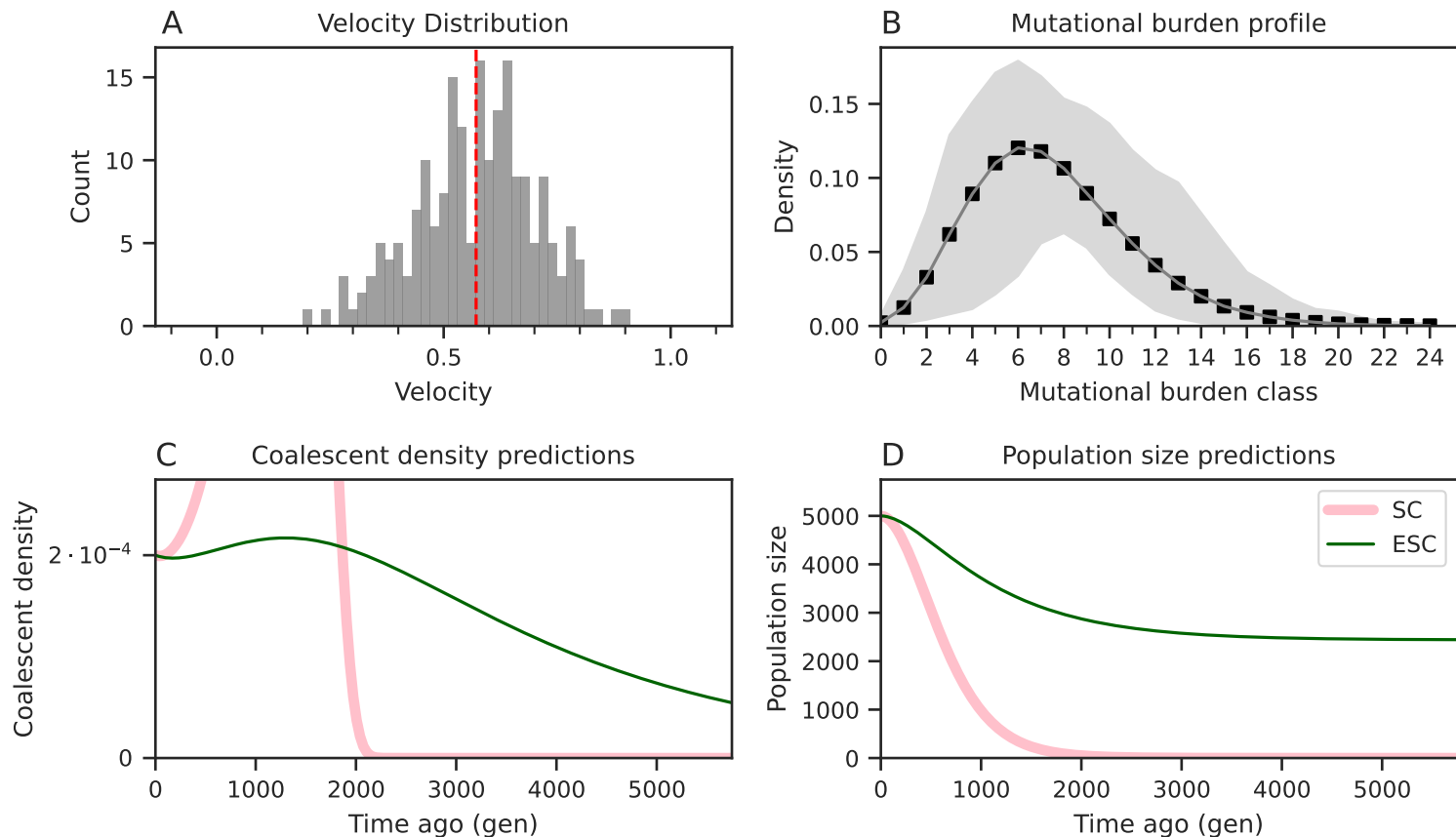
$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 4 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.6341 ± 0.1502 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 3.6 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity= 0.5982 ± 0.1500 , $n = 200$ simulations

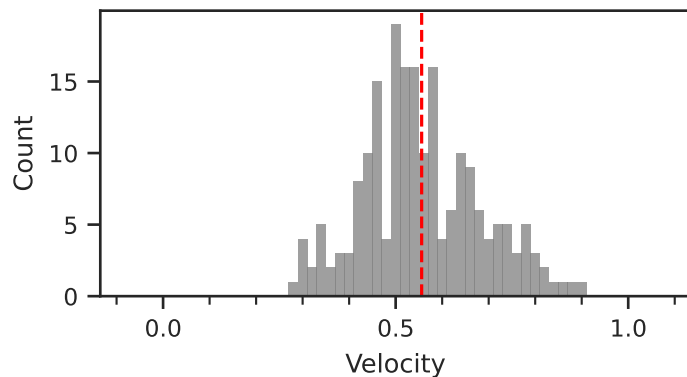


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 3.2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5714 ± 0.1313 , $n = 200$ simulations

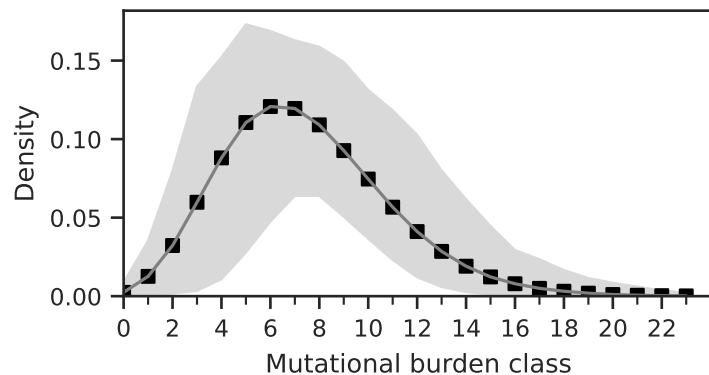


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2.8 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity= 0.5559 ± 0.1275 , $n = 200$ simulations

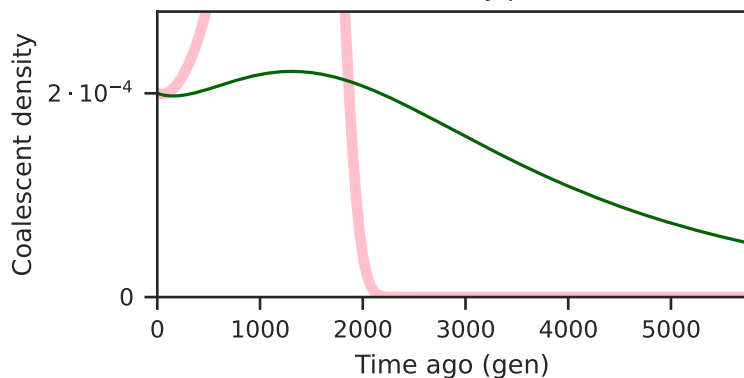
A Velocity Distribution



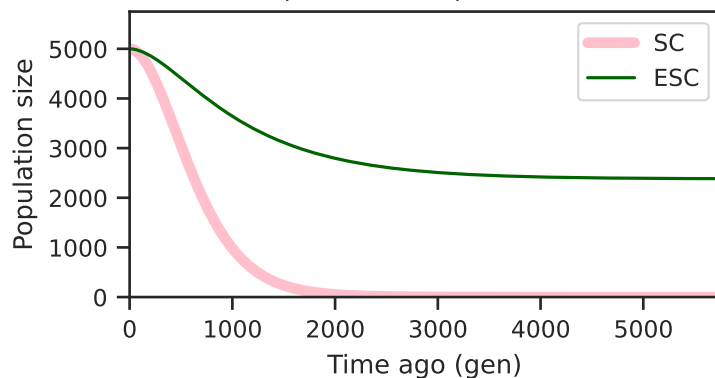
B Mutational burden profile



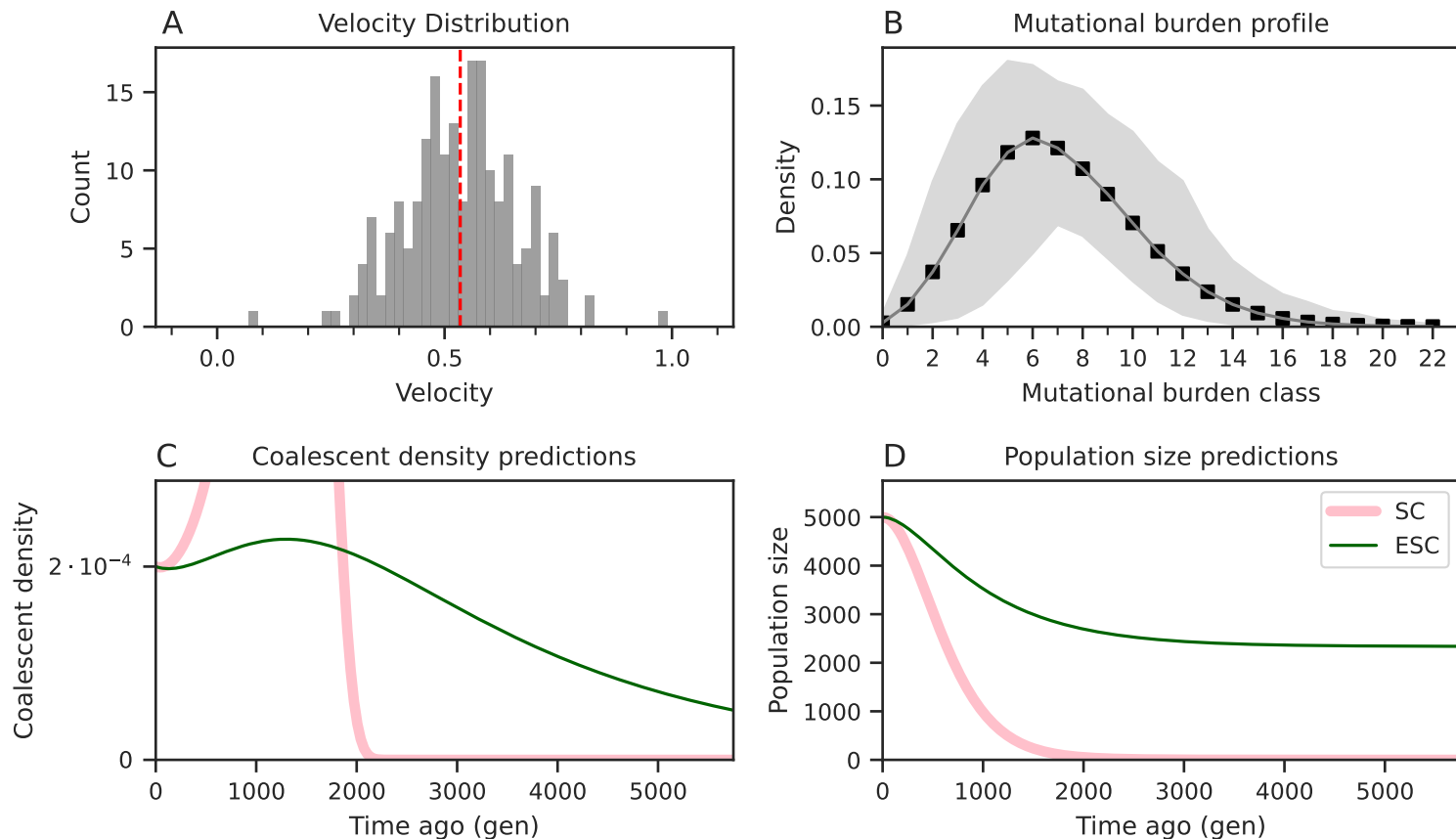
C Coalescent density predictions



D Population size predictions

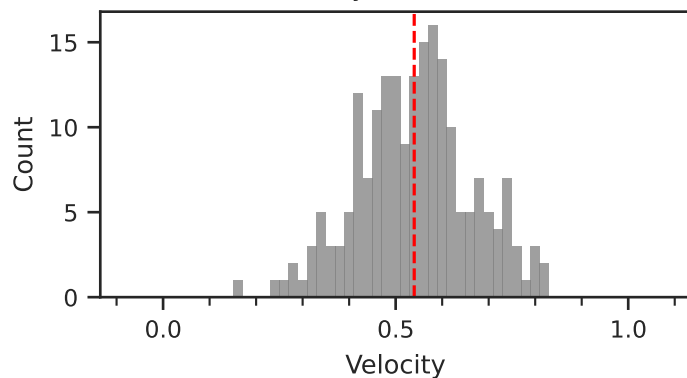


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2.4 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5345 ± 0.1243 , $n = 200$ simulations

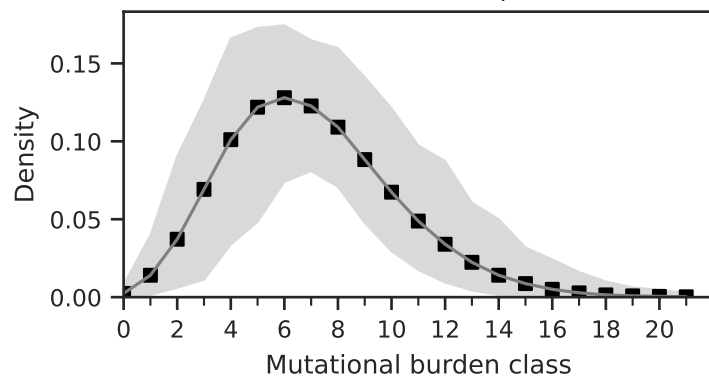


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5399 ± 0.1225 , $n = 200$ simulations

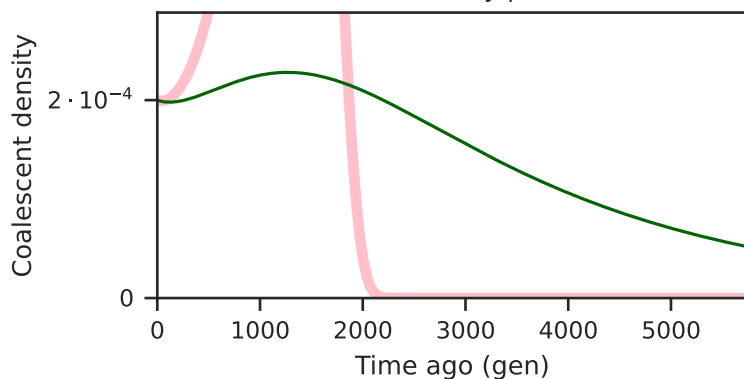
A Velocity Distribution



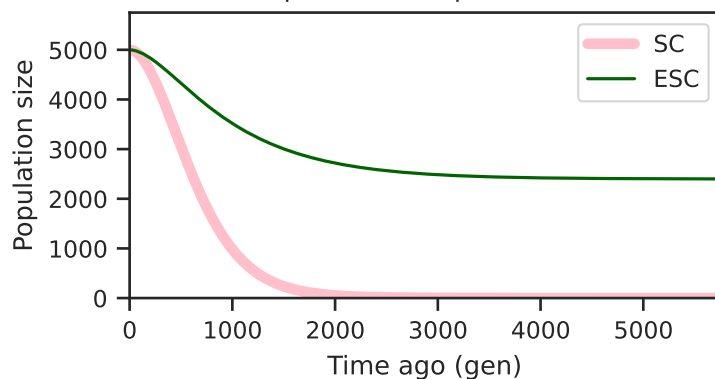
B Mutational burden profile



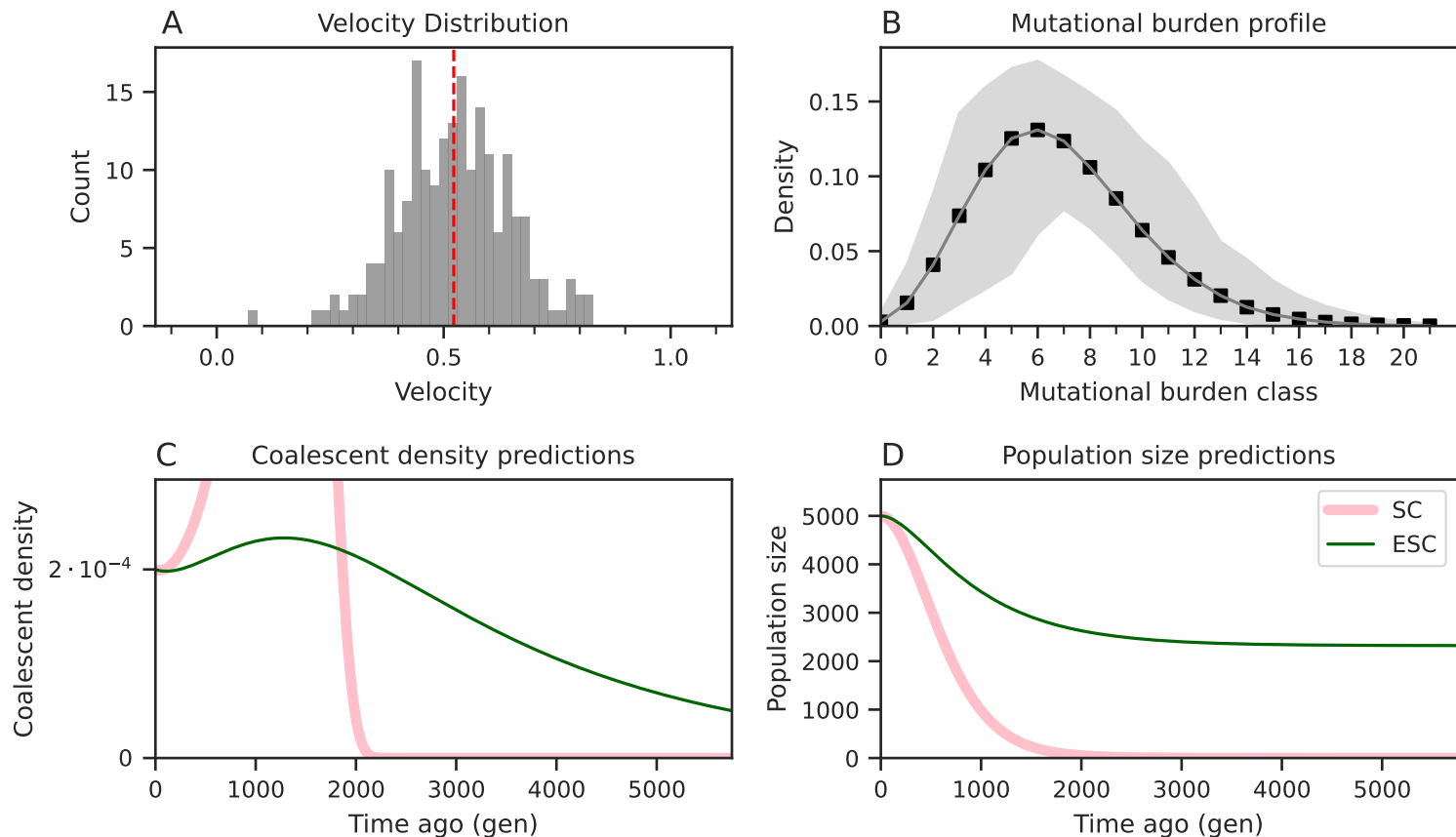
C Coalescent density predictions



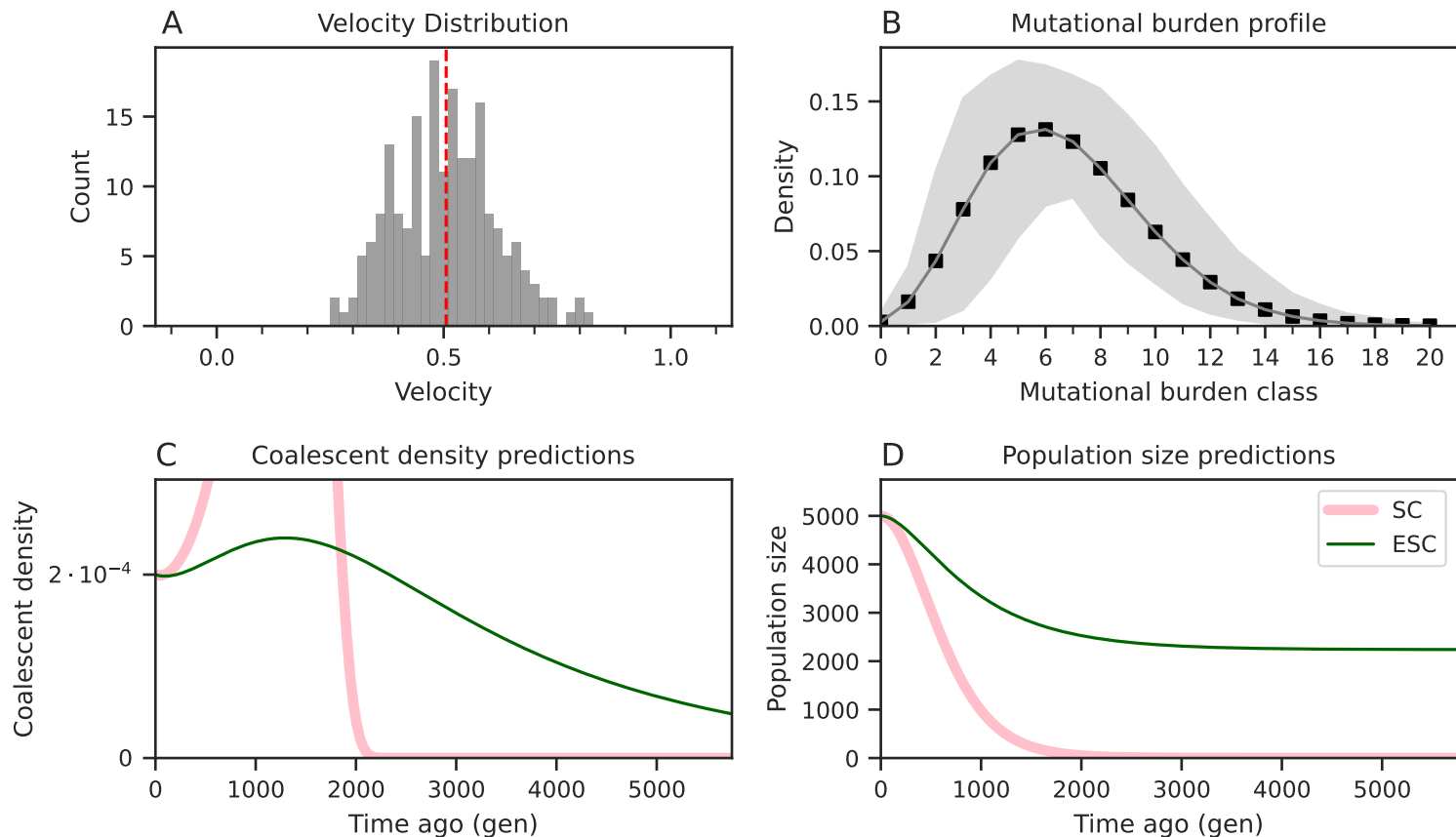
D Population size predictions



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1.6 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5222 ± 0.1233 , $n = 200$ simulations

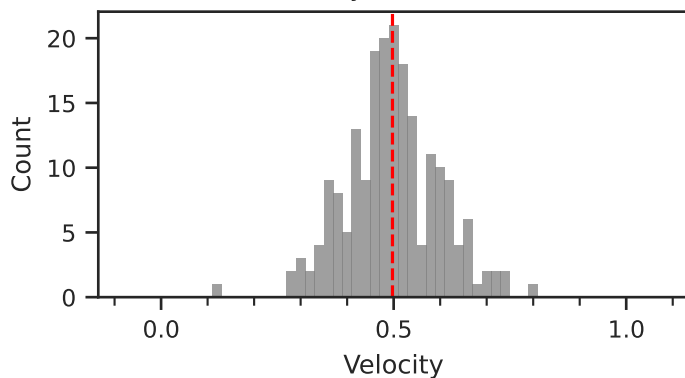


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1.2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5055 ± 0.1127 , $n = 200$ simulations

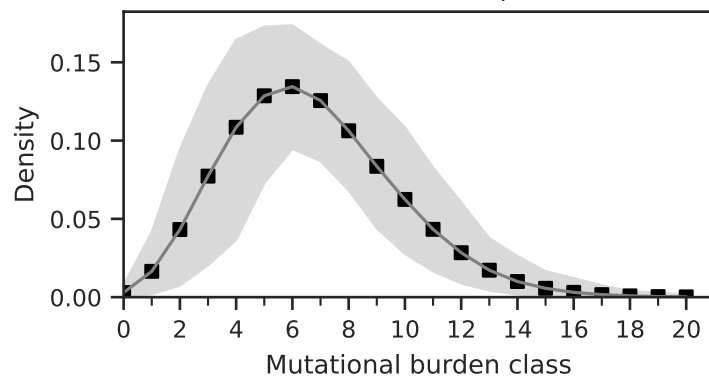


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 8 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.4975 ± 0.1001 , $n = 200$ simulations

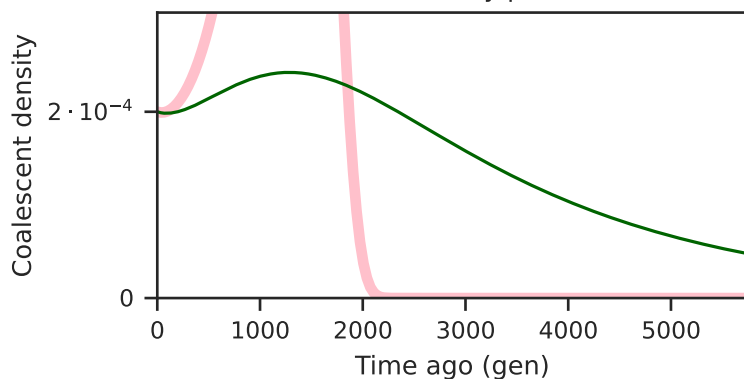
A Velocity Distribution



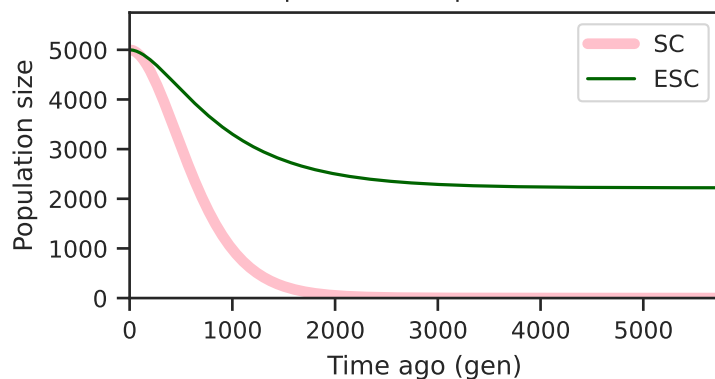
B Mutational burden profile



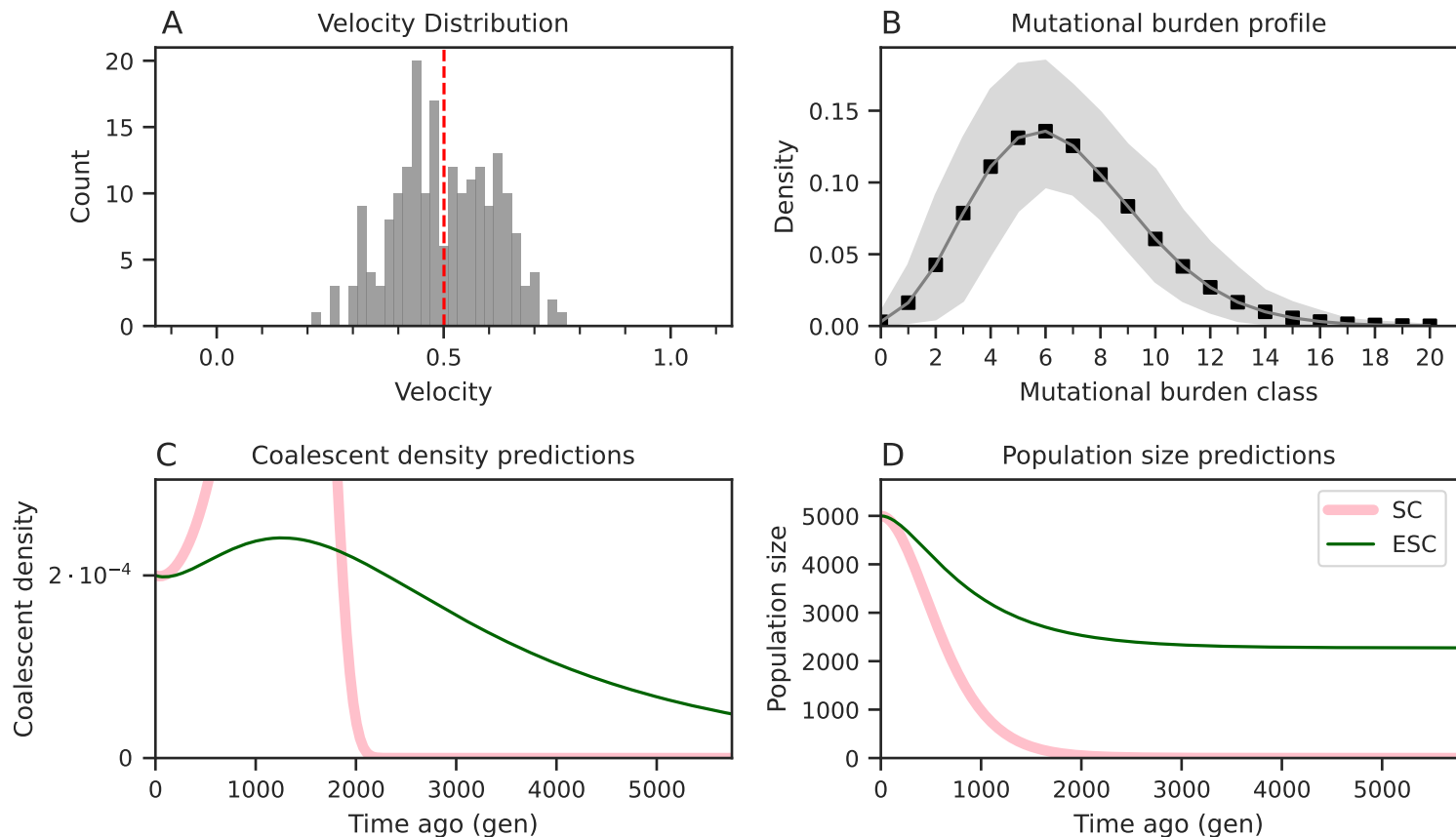
C Coalescent density predictions



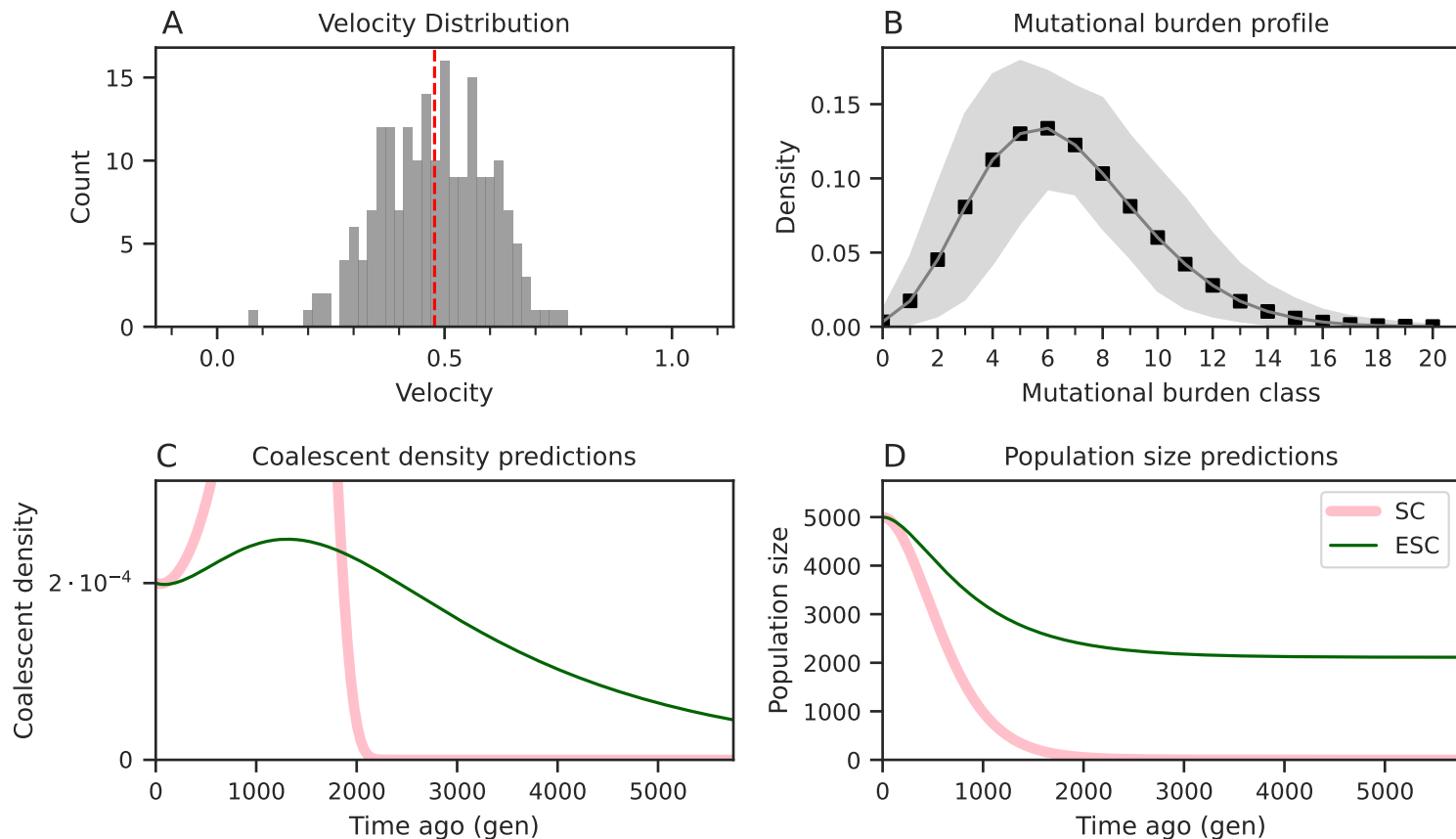
D Population size predictions



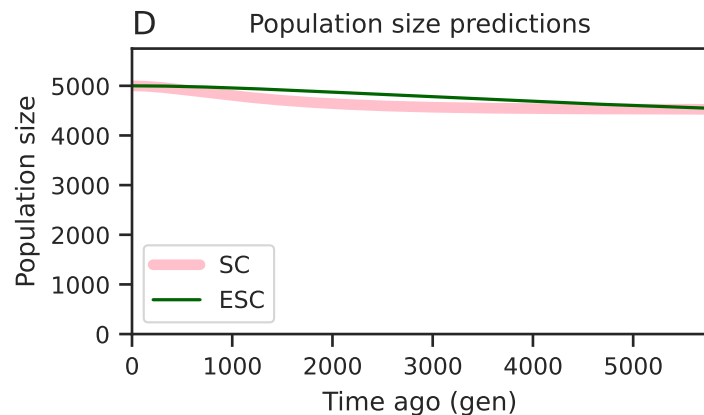
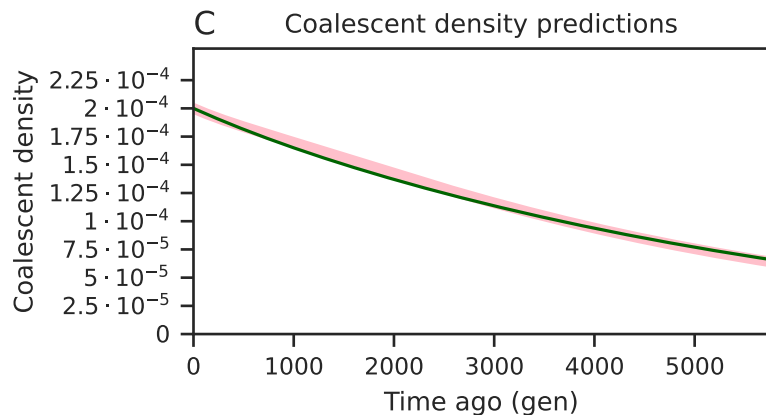
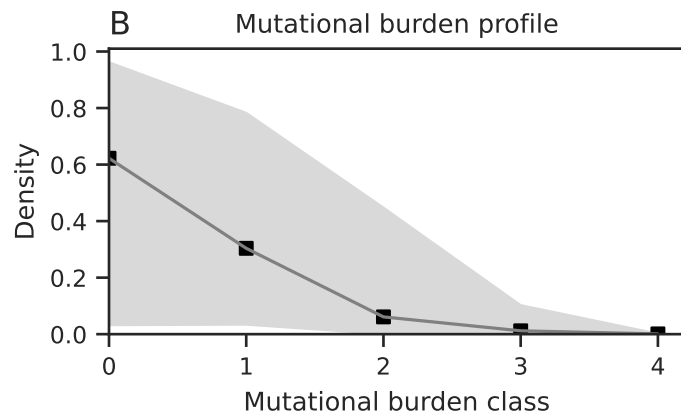
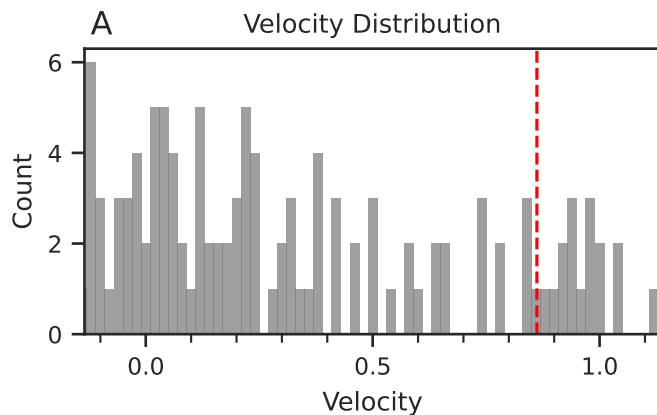
$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 4 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5007 ± 0.1108 , $n = 200$ simulations



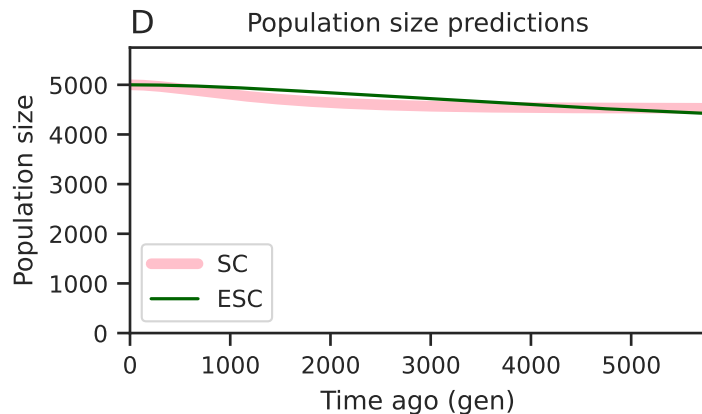
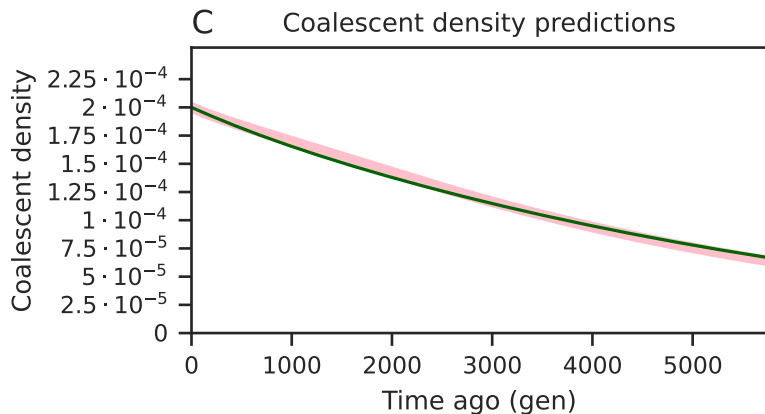
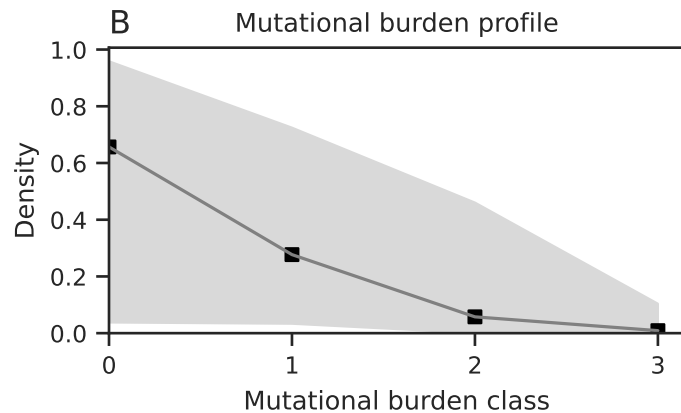
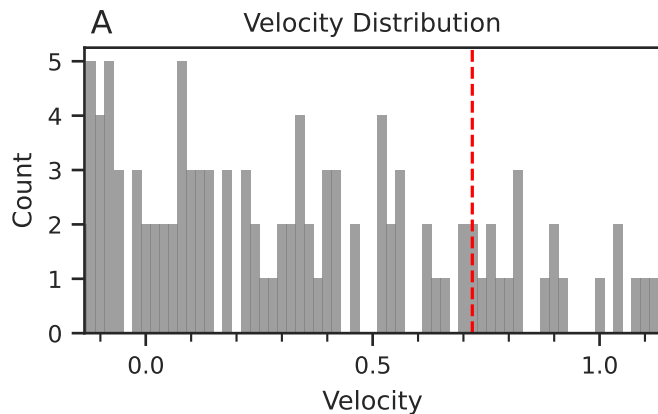
$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 0$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.4780 ± 0.1177 , $n = 200$ simulations



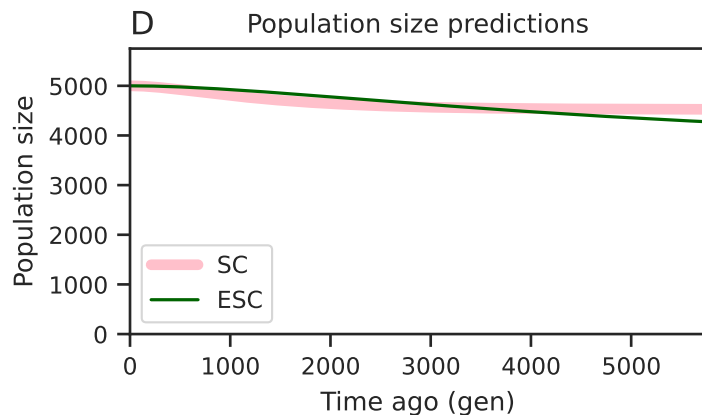
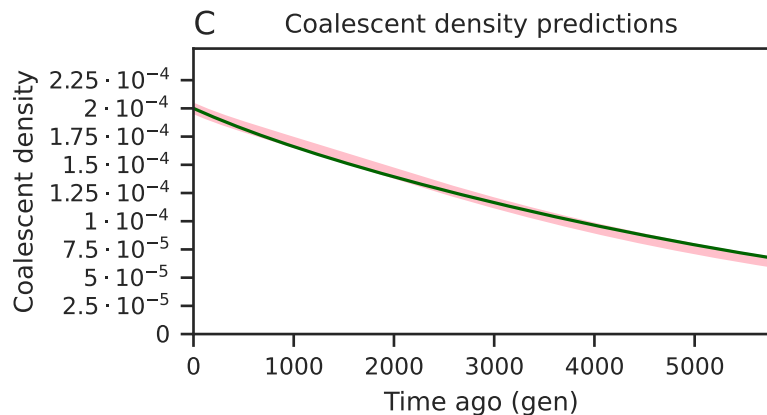
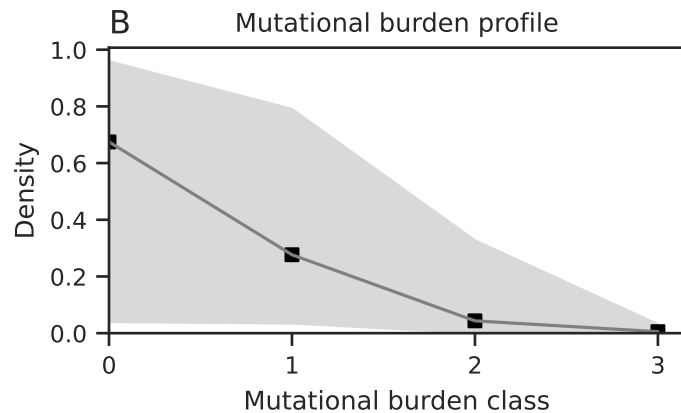
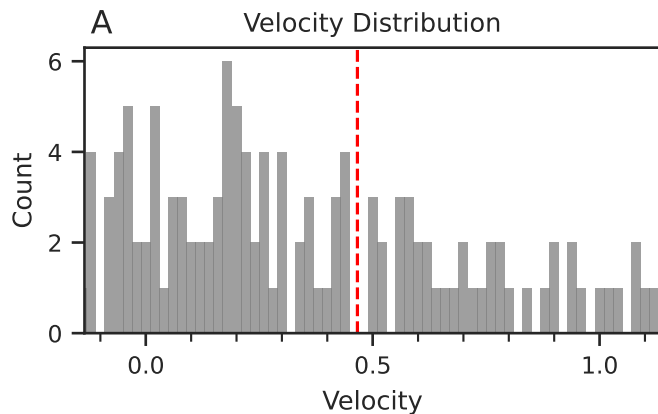
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 0.001$, $\phi = 4.52$
Mean Velocity= 0.8619 ± 1.2726 , $n = 200$ simulations



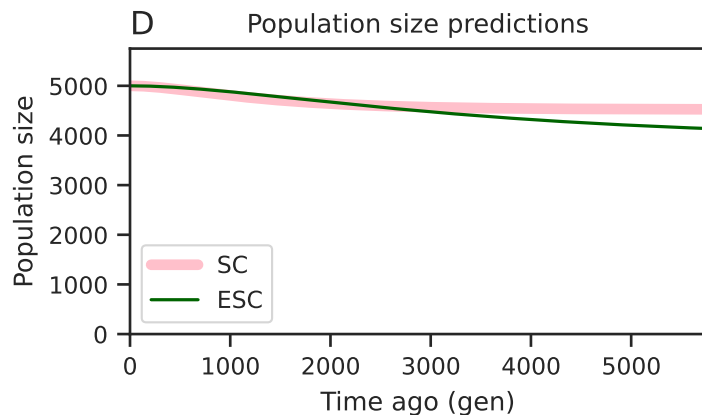
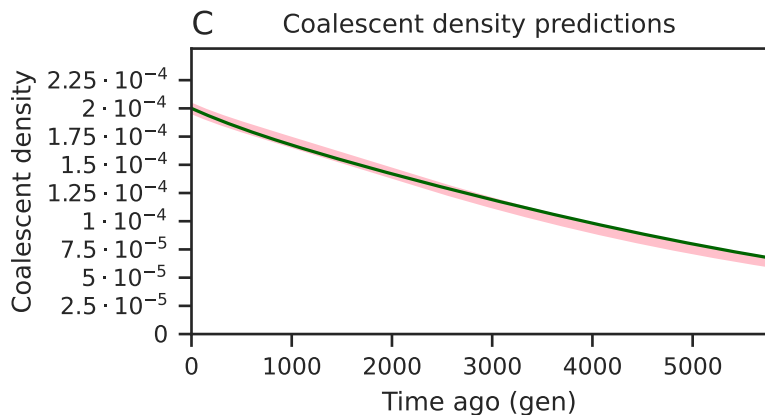
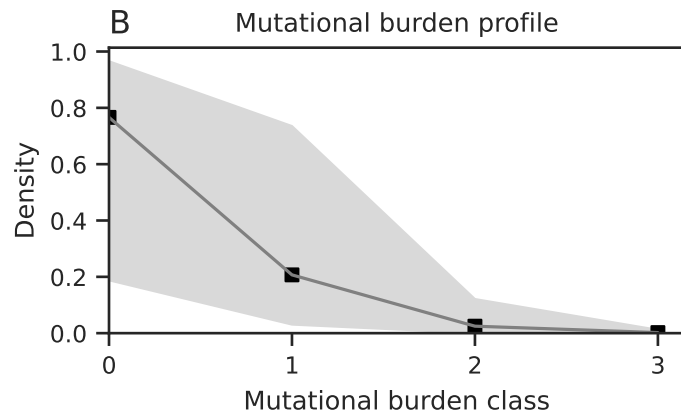
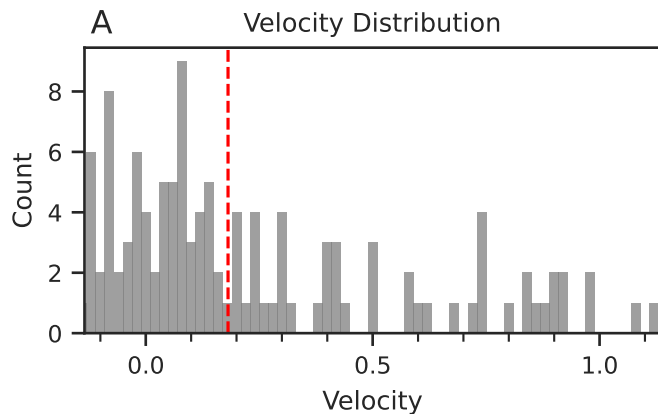
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 9 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity= 0.7196 ± 1.1174 , $n = 200$ simulations



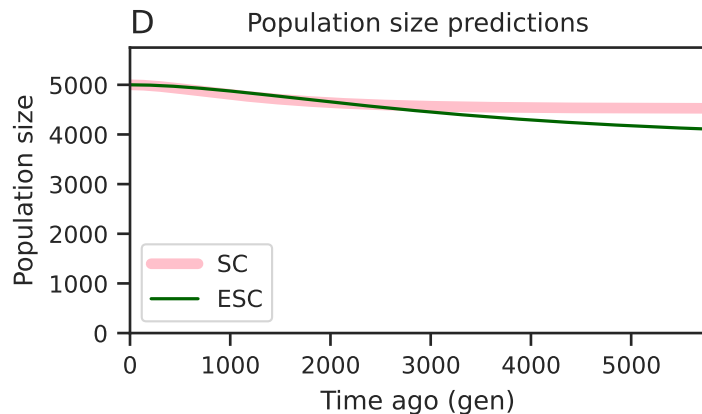
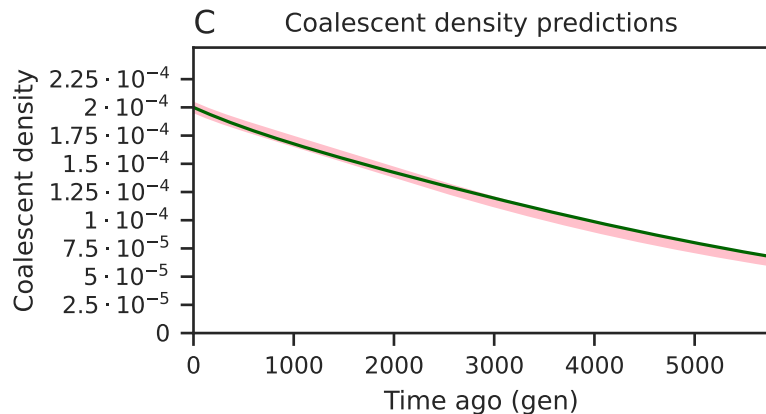
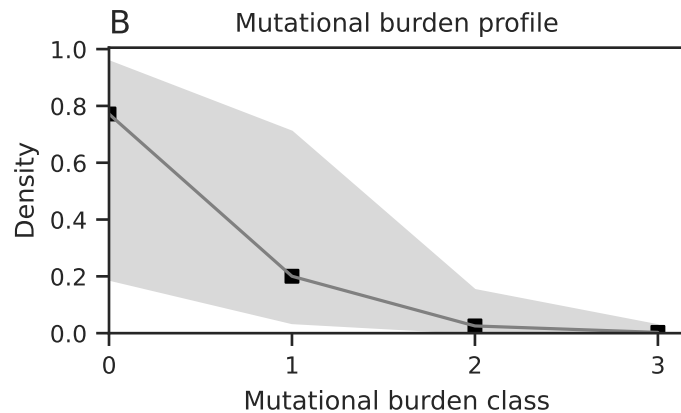
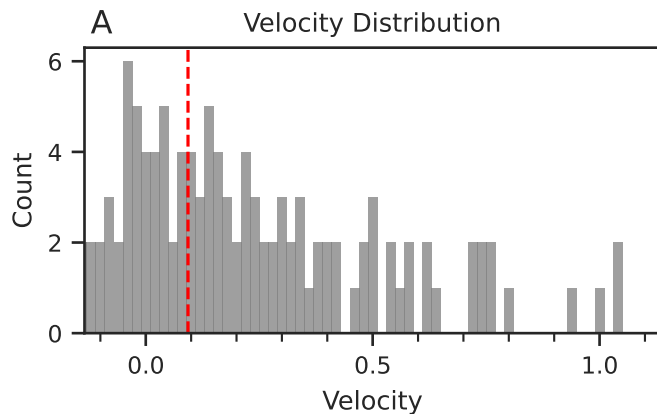
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 8 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity= 0.4664 ± 0.9524 , $n = 200$ simulations



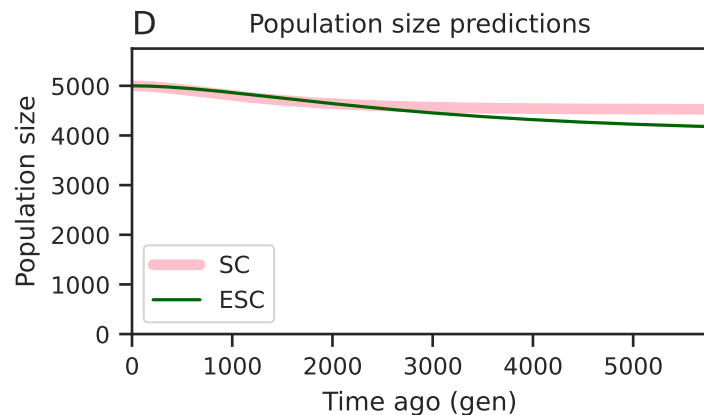
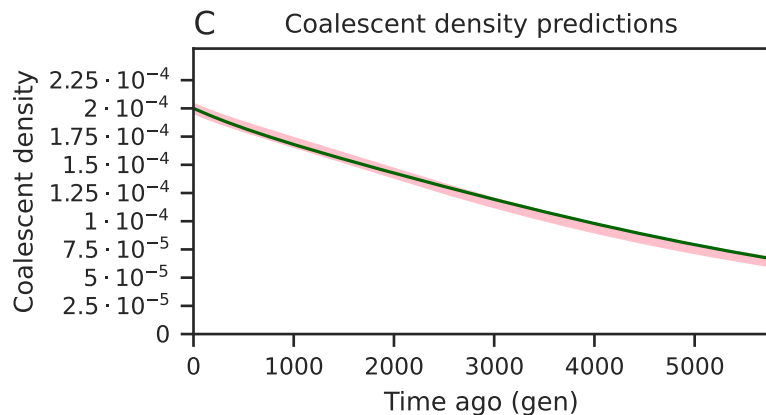
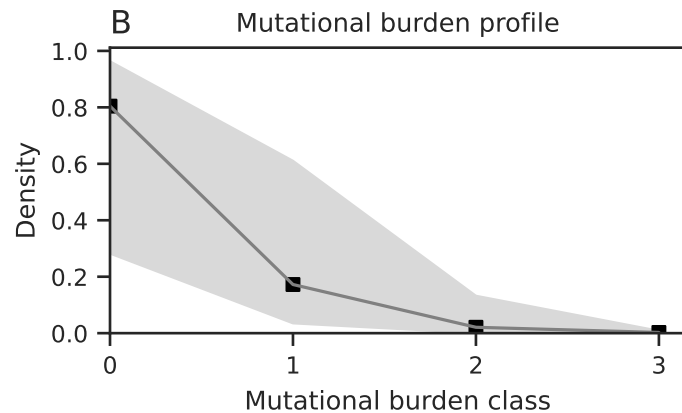
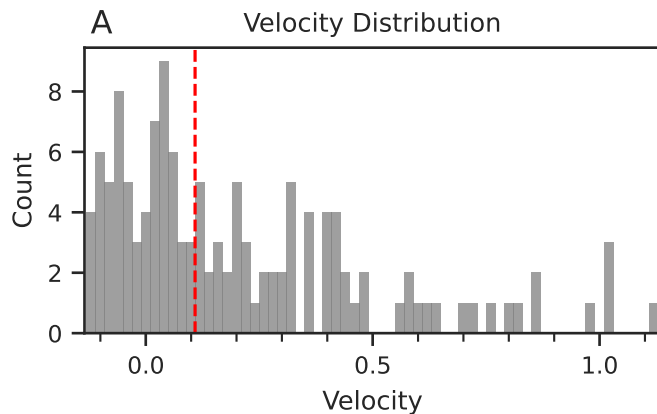
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 7 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity = 0.1813 ± 0.7899 , $n = 200$ simulations



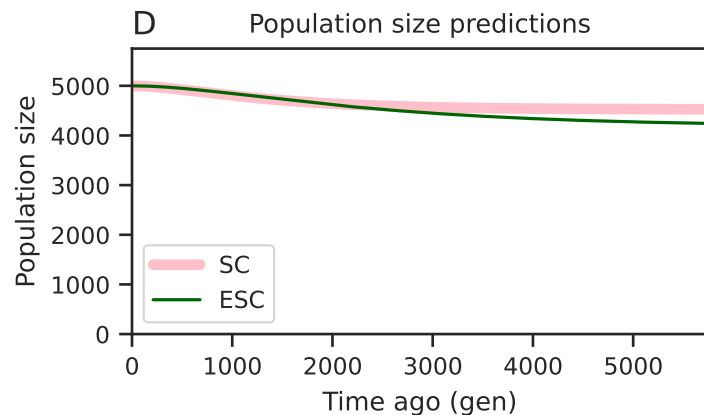
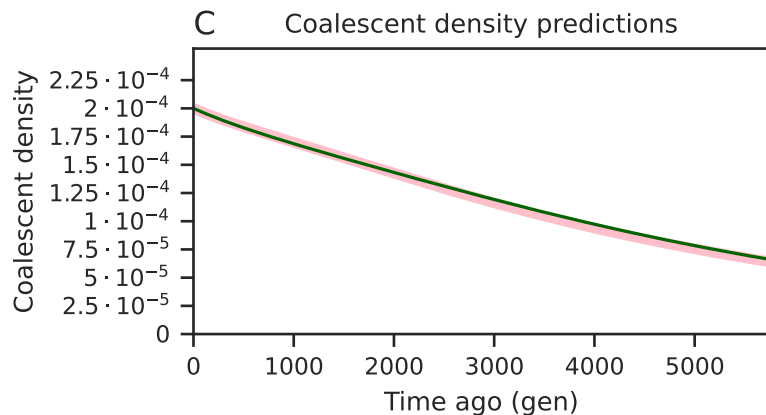
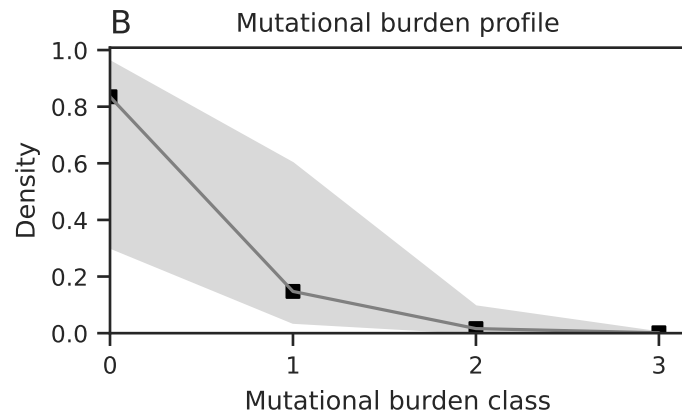
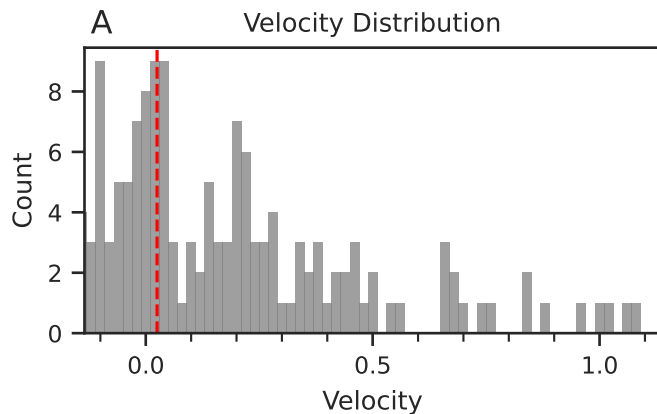
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 6 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity= 0.0931 ± 0.7895 , $n = 200$ simulations



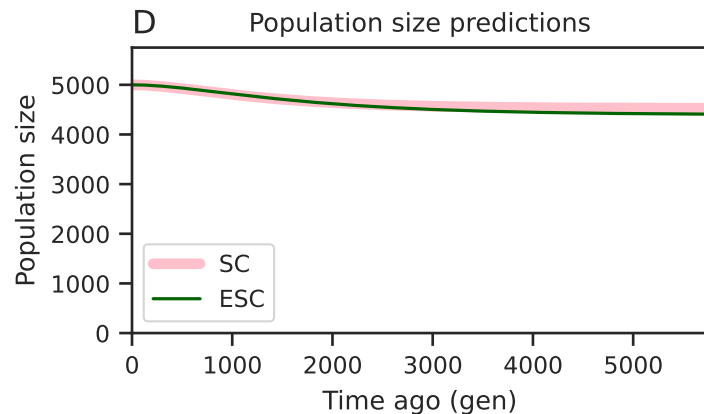
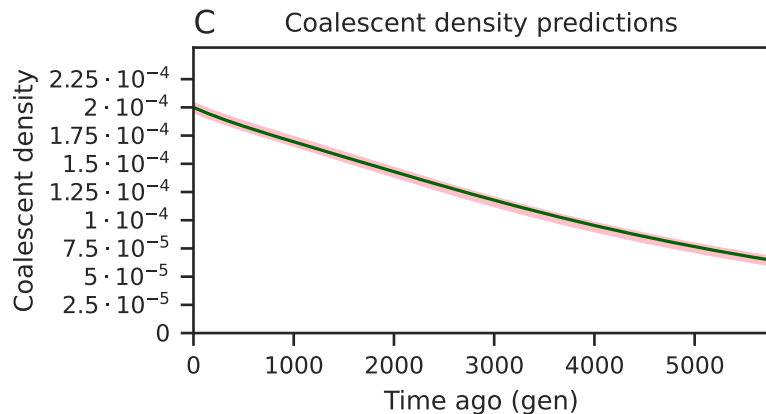
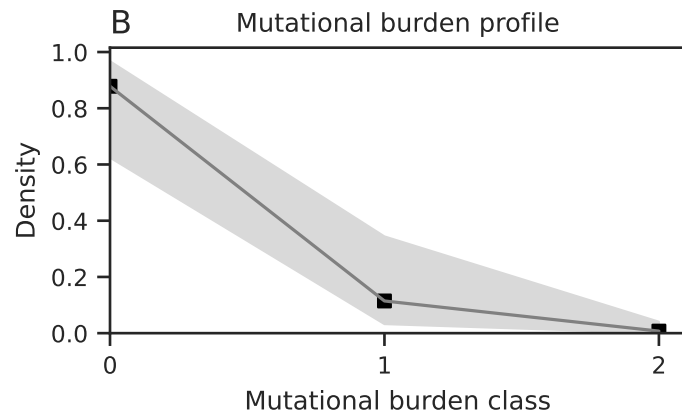
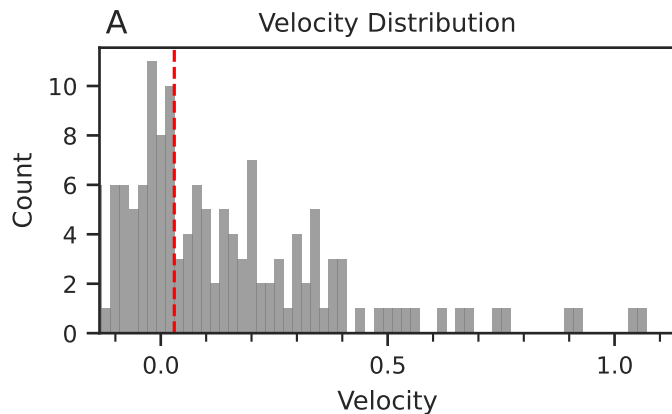
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity= 0.1087 ± 0.6616 , $n = 200$ simulations



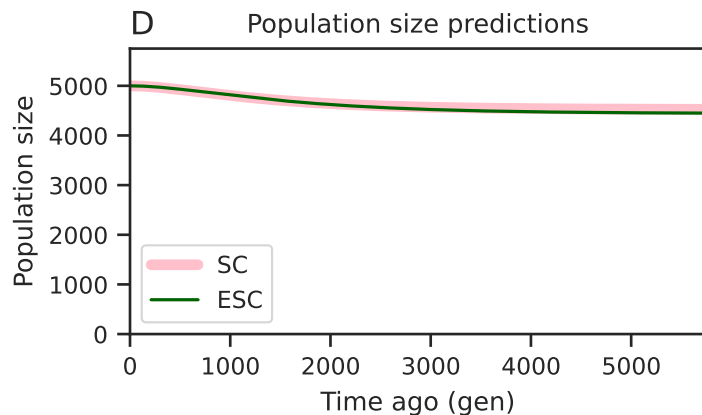
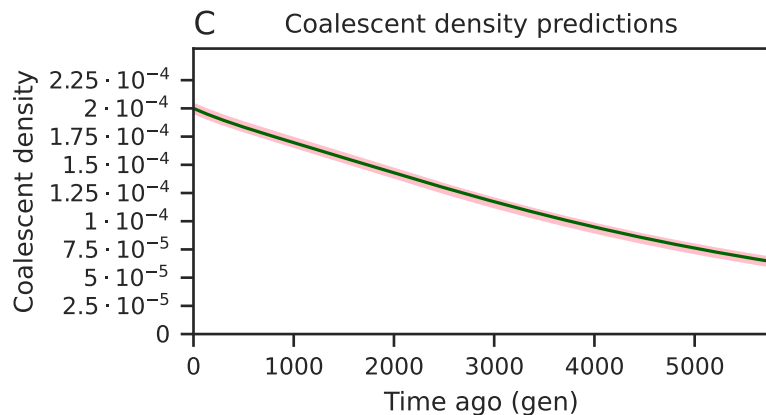
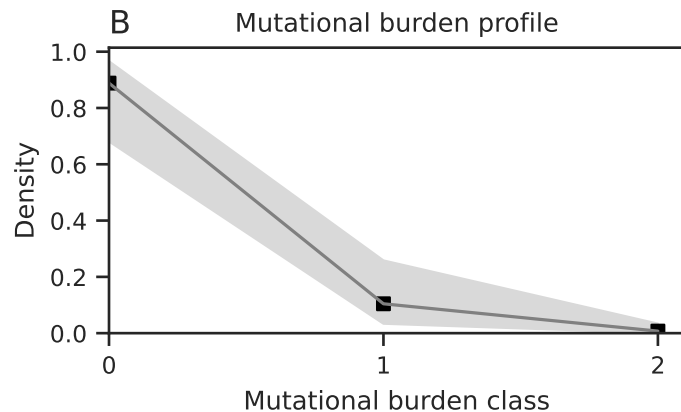
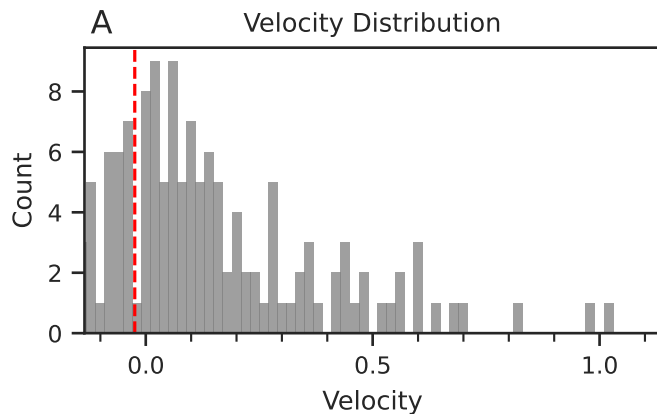
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 4 \cdot 10^{-4}$, $\phi = 4.52$
 Mean Velocity = 0.0249 ± 0.4753 , $n = 200$ simulations



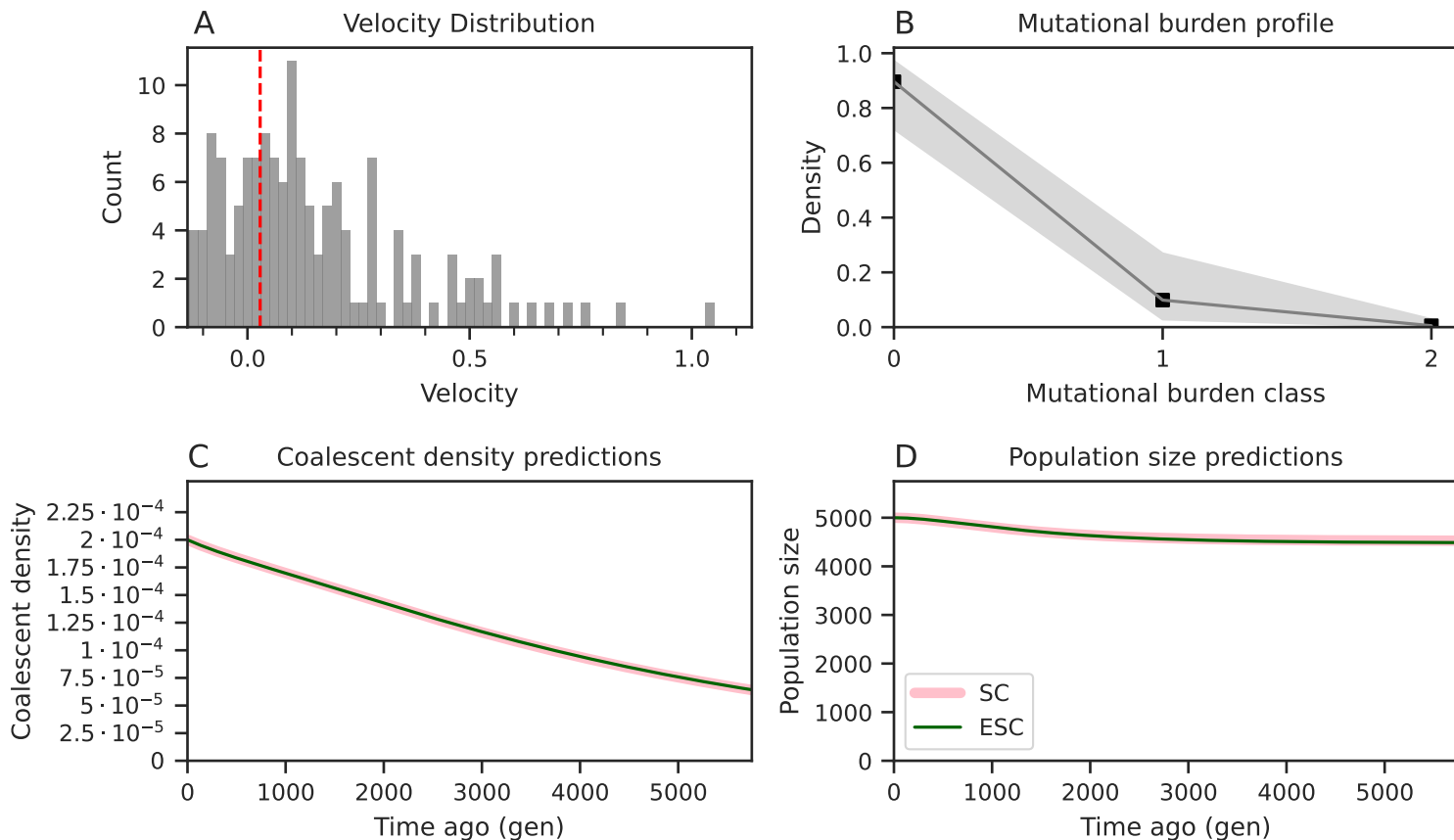
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity= 0.0298 ± 0.4354 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity = -0.0242 ± 0.3777 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity= 0.0285 ± 0.3554 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\sigma = 0$, $\phi = 4.52$
Mean Velocity = 0.0120 ± 0.3688 , $n = 200$ simulations

